

Affine and $\mathbb{P}$ rojective Geometry II

FUNG SAN GAAN

Affine and $\mathbb{P}$ rojective Geometry II

FUNG SAN GAAN

1	Week 1	3	8	Week 8 (empty)	20
2	Week 2	7	9	Week 9 (empty)	20
3	Week 3	9	10	Week 10	21
4	Week 4	13	11	Week 11	25
5	Week 5	17	12	Week 12	29
6	Week 6 (empty)	20	13	Appendix	30
7	Week 7 (empty)	20		Index	

These notes are adapted from professor J. B. Heller's lecture, FA25.

If you spot any error in these notes, please reach out to me via email at b192.168.1.91.03@gmail.com, or send me a direct message on Discord at [3fsg](#).

1 Week 1

8/25

We start with solution sets to polynomials $f_i \in k[x_1, \dots, x_n]$, where k is an algebraically closed field. The solution set looks like

$$V(f_i) = \{\vec{x} \in k^n \mid f_i(\vec{x}) = 0\}$$

Example. $C = \{Y^2 - X^3 - X^2 = 0\}$. Over the reals, this looks like a curve with a singular point.

The solution set itself does not see any geometry. So we need to put a topology on it. Recall that we used the Zariski topology on k^n . Given M set of polynomials from $k[x_1, \dots, x_n]$. The "vanishing locus" is

$$k^n \supseteq V(M) = \{\vec{x} \in k^n \mid f_i(\vec{x}) = 0, f \in M\}$$

Closed sets in Zariski topology on k^n are defined to be $V(I)$, where $I \subseteq k[x_1, \dots, x_n]$ is an ideal. We write $\mathbb{A}^n(k)$ for this space. Now we go over some basic results.

Vocabulary 1.1 (affine algebraic sets). They are closed subspaces of $\mathbb{A}^n(k)$.

Theorem 1 (Hilbert's Nullstellensatz). Let k be an algebraically closed field. Write $A = k[x_1, \dots, x_n]$. Let $I \subseteq A$ be an ideal. And let $f \in A$ such that $f(\vec{x}) = 0$ for all $x \in V(I)$. Then $f^r \in I$ for some r . $\square$

Corollary. There is a bijection

$$\{\text{algebraic subsets of } \mathbb{A}^n(k)\} \longleftrightarrow \{\text{radical ideals of } k[x_1, \dots, x_n]\}$$

Recall that $I \subseteq R$ is radical if $r^i \in I$ implies $r \in I$. $\square$

Vocabulary 1.2 (morphisms of algebraic sets). Let $X \subseteq \mathbb{A}^m(k)$ and $Y \subseteq \mathbb{A}^n(k)$ be algebraic sets. Morphisms between them are polynomial functions $f : X \rightarrow Y$. It is a map of sets such that there are polynomials $f_1, \dots, f_n \in k[x_1, \dots, x_m]$ such that $f(\vec{x}) = (f_1(\vec{x}), \dots, f_n(\vec{x}))$.

Example. Any polynomial $f \in k[x_1, \dots, x_n]$ gives us a map $\mathbb{A}^n(k) \rightarrow \mathbb{A}^1(k)$. More generally, if $X \subseteq \mathbb{A}^n(k)$ is an algebraic subset, then $f \in k[x_1, \dots, x_n]$ defines $f : X \rightarrow \mathbb{A}^1(k)$.

Example. Suppose we have a map $\mathbb{A}^1(k) \rightarrow C = \{Y^2 - X^3 - X^2 = 0\} \subseteq \mathbb{A}^2(k)$ given by $x \mapsto (x^2 - 1, x^3 - x)$. This map is only a bijection when $\text{char}(k) = 2$. In that case, $\pm 1 \mapsto (0, 0)$, but it is not an isomorphism.

Example. The exponential map $\exp : \mathbb{A}^1(\mathbb{C}) \rightarrow \mathbb{A}^1(\mathbb{C})$ is *not* algebraic.

Observe that $\text{hom}(X, \mathbb{A}^1(k))$ is a ring. Operations are $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$.

Vocabulary 1.3 (affine coordinate rings). Let $X \subseteq \mathbb{A}^n(k)$ be an algebraic set. Its coordinate ring is $\Gamma(X) = \text{hom}(X, \mathbb{A}^1(k))$. Observe that we have a surjection $k[x_1, \dots, x_n] \twoheadrightarrow \Gamma(X)$ and $I(X) = \ker(k[x_1, \dots, x_n] \twoheadrightarrow \Gamma(X))$. So

$$\Gamma(X) \simeq k[x_1, \dots, x_n]/I(X)$$

By the Nullstellensatz, $I(X)$ is radical implies $\Gamma(X)$ is reduced. Recall that reduced means no nilpotents. Now we have the assignment

$$\{\text{affine alg sets}\} \xrightarrow{\simeq} \{\text{reduced finitely generated } k \text{ alg}\}$$

given by $X \mapsto \Gamma(X)$. This is a bijection. But even better,

$$\text{hom}_{\text{alg set}}(X, Y) \simeq \text{hom}_{k\text{-alg}}(\Gamma(Y), \Gamma(X))$$

Let $I \subseteq \Gamma(X)$ be an ideal. The topology on X is the one whose closed sets are $V(I) = \{x \in X \mid f(x) = 0, \forall f \in I\}$. Consider $x = (x_1, \dots, x_n) \in X \subseteq \mathbb{A}^n(k)$. The ideal $\mathfrak{m}_x = \{f \in \Gamma(X) \mid f(x) = 0\} = \ker(\text{ev}_x : \Gamma(X) \rightarrow k)$ is maximal because the resulting quotient is a field. In fact, one can check every maximal ideal of $\Gamma(X)$ is of the form $\mathfrak{m}_x$ for some x . In particular, maximal ideals of $\Gamma(\mathbb{A}^n(k)) = k[x_1, \dots, x_n]$ are of the form $(x_1 - a_1, \dots, x_n - a_n)$ for some $(a_1, \dots, a_n) \in k^n$. As a set,

$$X = \{\text{maximal ideals of } \Gamma(X)\} \quad V(I) = \{\mathfrak{m} \subseteq \Gamma(X) \mid I \subseteq \mathfrak{m}\}$$

8/27

Vocabulary 1.4 (poset of opens). Let X be a topological space. $\text{Op}(X)$ is the poset of opens $U \subseteq X$. And $U \subseteq V$ corresponds to a morphism $U \rightarrow V$ in $\text{Op}(X)$.

Vocabulary 1.5 (presheaf). A presheaf of a set (Ab , RMod , or some ring) is a contravariant functor $\mathcal{F} : \text{Op}(X)^{\text{op}} \rightarrow \text{Set}$. Concretely, a presheaf $\mathcal{F}$ of abelian groups on X consists of the data

- (a) for every open subset $U \subseteq X$, an abelian group $\mathcal{F}(U)$, and
- (b) for every inclusion $V \subseteq U$ of open subsets of X , a morphism of abelian groups $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$,

subject to the conditions

- (0) $\mathcal{F}(\emptyset) = 0$, where $\emptyset$ is the empty set,
- (1) ρ_{UU} is the identity map $\mathcal{F}(U) \rightarrow \mathcal{F}(U)$, and
- (2) if $W \subseteq V \subseteq U$ are three open subsets, then $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$.

Example. Here are some simple examples of presheaves.

1. Let $A \in \text{Ab}$. The constant presheaf valued at A is $cA : \text{Op}(X)^{\text{op}} \rightarrow \text{Ab}$. For *any* open $U \subseteq X$, $cA(U) = A$.
2. Let $U \subseteq X$ be any open subset. Recall that $R_U = \text{hom}_{\text{cont}}(U, \mathbb{R})$ is a ring. The assignment $\mathcal{C} : \text{Op}(X)^{\text{op}} \rightarrow \text{Ring}$ given by $\mathcal{C}(U) = R_U$ is called the presheaf of $\mathbb{R}$ -valued functions on X . If $U \subseteq V$,

$$\mathcal{C}(V) = \text{hom}(V, \mathbb{R}) \longrightarrow \mathcal{C}(U) = \text{hom}(U, \mathbb{R})$$

$$f : V \rightarrow \mathbb{R} \longmapsto f|_U : U \subseteq V \rightarrow \mathbb{R}$$

Vocabulary 1.6 (sheaf). A sheaf of sets on X is a presheaf $\mathcal{F}$ on X such that for any open cover $\{U_i\}$ of some open $U \subseteq X$, the diagram is exact:

$$\mathcal{F}(U) \xrightarrow{\gamma} \prod_i \mathcal{F}(U_i) \xrightarrow[\sigma']{\sigma} \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

$$s \longmapsto (s|_{U_i} := \rho_{UU_i}(s))_i \xrightarrow[\quad]{(s_i|_{U_i \cap U_j})_{i,j}} (s_j|_{U_i \cap U_j})_{i,j}$$

The exactness here means γ is injective and its image consists of $(s_i)_i \in \prod_i \mathcal{F}(U_i)$. If $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there is an element $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for each i . Concretely, $\mathcal{F}$ is sheaf if it is a presheaf and satisfies the following additional conditions

- (3) if U is an open set, if $\{V_i\}$ is an open covering of U , and if $s \in \mathcal{F}(U)$ is an element such that $s|_{V_i} = 0$ for all i , then $s = 0$;
- (4) if U is an open set, if $\{V_i\}$ is an open covering of U , and if we have elements $s_i \in \mathcal{F}(V_i)$ for each i , with the property that for each i, j ,

$$s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j},$$

then there is an element $s \in \mathcal{F}(U)$ such that $s|_{V_i} = s_i$ for each i .

Vocabulary 1.7 (category of $\text{Psh}(X)$). Let $\mathcal{F}$ and $\mathcal{G}$ be two presheaves of sets on X . A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ consists of a family of morphisms $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ of sets for each open set U of X , subject to the condition that this morphism is compatible with restrictions. For all $U \subseteq V$ open, the following diagram commutes

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \\ \rho_{VU} \downarrow & & \rho_{VU} \downarrow \\ \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \end{array}$$

Morphisms of presheaves are morphism of sheaves. In particular, $\text{Sh}(X) \subseteq \text{Psh}(X)$ is a full subcategory.

Example. Examples and nonexamples of sheaves.

1. The constant presheaf cA is usually not a sheaf. For example, if $X = \{a, b\}$ with discrete topology, then $X = \{a\} \cup \{b\}$ is an open cover. Let $s_{\{a\}} \in \mathcal{F}(\{a\})$ and $s_{\{b\}} \in \mathcal{F}(\{b\})$. For this open cover, $s_{\{a\}}|_{\{a\} \cap \{b\}}$ translates to a map $\rho_{\{a\}, \{a\} \cap \{b\}}(s_{\{a\}}) : \mathcal{F}(\{a\}) \rightarrow \mathcal{F}(\emptyset)$ that is given by $s_{\{a\}} \mapsto \{p\}$, where $\{p\}$ is a point in $\mathcal{F}(X)$. Similarly, $s_{\{b\}}|_{\{a\} \cap \{b\}}$ is the map $\mathcal{F}(\{b\}) \rightarrow \mathcal{F}(\emptyset)$ given by $s_{\{b\}} \mapsto \{p'\}$. Since restriction maps are well-defined, $\{p\} = \{p'\}$. Now the sheaf condition guarantees two maps $s|_{\{a\}} = \rho_{X\{a\}}(s) = s_{\{a\}}$ and $s|_{\{b\}} = \rho_{X\{b\}}(s) = s_{\{b\}}$. These maps are $\mathcal{F}(X) \rightarrow \mathcal{F}(\{a\})$ and $\mathcal{F}(X) \rightarrow \mathcal{F}(\{b\})$ respectively. Then $\mathcal{F}(X) \subseteq \mathcal{F}(\{a\}) \times \mathcal{F}(\{b\})$. And if $(s_{\{a\}}, s_{\{b\}}) \in \mathcal{F}(\{a\}) \times \mathcal{F}(\{b\})$, restriction maps take them to $(\{p\}, \{p\}) \in \mathcal{F}(X)$. So $\mathcal{F}(X) = \mathcal{F}(\{a\}) \times \mathcal{F}(\{b\})$. But $A = cA(X) \neq cA(a) \times cA(b) = A \times A$.

2. Let $\mathcal{C} : \mathbf{Op}(X)^{\text{op}} \rightarrow \mathbf{Ring}$ be a sheaf of $\mathbb{R}$ -valued functions. Let $U = \bigcup_i U_i$ be an open cover. The sheaf condition is satisfied because

- (a) If $g : U \rightarrow \mathbb{R}$ satisfies $g|_{U_i} = 0$, then $g = 0$.
- (b) Given $f_i : U_i \rightarrow \mathbb{R}$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Then define $h : U \rightarrow \mathbb{R}$ such that

$$h(t) = \begin{cases} f_i(t) & \text{if } t \in U_i \\ f_j(t) & \text{if } t \in U_j \end{cases}$$

Vocabulary 1.8 (stalks). Let $\mathcal{F}$ be a presheaf and X is some space. Let $x \in X$. Write $\text{nbhd}(x) \subseteq \mathbf{Op}(x)$ as a subset of opens which contain $x \in X$. The stalk at x is defined as

$$\mathcal{F}_x := \text{colim}_{U \in \text{nbhd}(x)} \mathcal{F}(U) = \{(U, s) | x \in U, s \in \mathcal{F}(U)\} / \sim$$

The colimit can be described as $\mathcal{F}_x =$ equivalence classes of (U, s) , where $s \in \mathcal{F}(U)$. $(U, s) \sim (V, t)$ is equivalent if there is a neighborhood $W \subseteq U \cap V$ of x such that $s|_W = t|_W \in \mathcal{F}(W)$.

Example. Let $X = \mathbb{C}$ and $\mathcal{F}$ is the sheaf of analytic functions. Let $z_0 \in \mathbb{C}$ and f_1, f_2 on some neighborhood U of z_0 (i.e. $f_1, f_2 \in \mathcal{F}(U)$). These agree on some small neighborhood of z_0 if and only if f_1, f_2 have the same Taylor expansion. In other words,

$$\mathcal{F}_x = \left\{ \sum_{i \geq 0} a_n (z - z_0)^n \mid \text{power series with positive r.o.c.} \right\}$$

Q. What are stalks good for?

A. The "stalk at x " is a functor $\mathbf{Psh}(X) \rightarrow \mathbf{Set}$ given by $\mathcal{F} \mapsto \mathcal{F}_x$.

8/29

Proposition 1. Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be presheaves. And let $\varphi, \psi : \mathcal{F} \rightarrow \mathcal{G}$ be morphisms of presheaves. We have that

1. If $\mathcal{F}$ is a sheaf, $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective for all $x \in X$ if and only if $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective for all U .
2. If both $\mathcal{F}, \mathcal{G}$ are sheaves, then $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is a bijection for all $x \in X$ if and only if $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is a bijection for all $U \subseteq X$ open.
3. If both $\mathcal{F}, \mathcal{G}$ are sheaves, then $\varphi_x = \psi_x$ for all $x \in X$ if and only if $\varphi = \psi$.

Proof. We claim that if $\mathcal{F}$ is a sheaf, then the map $\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x$ given by $s \mapsto (s_{x_1}, \dots, s_{x_n}, \dots)$ is injective. Here each x_i is in U and they do not have to be countable or finite. On each coordinate, we have a canonical map $\mathcal{F}(U) \rightarrow \mathcal{F}_{x_i}$ that takes s to the equivalence class $[(U, s)] \in \mathcal{F}_{x_i}$. We call s_{x_i} the germ of s in x_i . To show it is injective, let $s, t \in \mathcal{F}(U)$, and assume $(s_{x_i})_{x_i \in U} = (t_{x_i})_{x_i \in U}$ for all x_i . In other words, $[(U, s)] = [(U, t)] \in \mathcal{F}_{x_i}$ for all x_i . This implies for each x_i , there exists a neighborhood $V_{x_i} \subseteq U$ of x_i such that $s|_{V_{x_i}} = t|_{V_{x_i}}$. Now form an open cover $U = \bigcup_{x \in U} V_x$. The sheaf condition says that $s = t$. Consider the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\alpha} & \prod_{x \in U} \mathcal{F}_x \\ \downarrow \varphi_U & & \downarrow \prod_x \varphi_x \\ \mathcal{G}(U) & \xrightarrow{\beta} & \prod_{x \in U} \mathcal{G}_x \end{array}$$

1. ($\Leftarrow$) this follows from a standard fact that filtered colimits of injections are injections. ($\Rightarrow$) since $\prod_x \varphi_x, \beta$, and α are injections, this would force φ_U to be an injection.
2. ($\Leftarrow$) immediate. ($\Rightarrow$) we need to show that if $\prod_x \varphi_x$ is bijective, then $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is bijective. Since (1.) already showed it is injective, it remains to show it is surjective. Let $t \in \mathcal{G}(U)$. We claim that for each $x \in U$, there is a neighborhood V_x and an element $s_{V_x} \in \mathcal{F}(V_x)$ such that $\varphi(s_{V_x}) = t|_{V_x}$. Indeed, since the induced map is defined by

$$\varphi_x := \text{colim}_{U \ni x} \varphi_U : \mathcal{F}_x \rightarrow \mathcal{G}_x$$

Once we viewed $\mathcal{F}_x$ as a colimit, we get a canonical map $\mathcal{F}(U) \rightarrow \mathcal{F}_x$ for each U . In other words, the following diagram commutes due to the universal property of colimit¹:

$$\begin{array}{ccc} \mathcal{F}(V_x) & \longrightarrow & \mathcal{F}_x \\ \downarrow & & \downarrow \\ \mathcal{G}(V_x) & \longrightarrow & \mathcal{G}_x \end{array} \quad \begin{array}{ccc} s & \longmapsto & s_x \\ \downarrow & & \downarrow \\ \varphi(s) & \longmapsto & t_x \end{array}$$

Note that both $t|_{V_x}$ and $\varphi(s_{V_x})$ get mapped to t_x . This means $[(V_x, \varphi(s_{V_x}))] = [(V_x, s_{V_x})]$. By 1.8, $\varphi(s_{V_x}) = t|_{V_x}$ on a small enough neighborhood. Since this is true for every $x \in U$, we get an open cover $U = \bigcup_x V_x$. We claim that $\{s_{V_x}\}$ glue to some element $s \in \mathcal{F}(U)$. To see this, we need to check $s_{V_x}|_{V_x \cap V_y} = s_{V_y}|_{V_x \cap V_y}$:

$$t|_{V_x \cap V_y} = \varphi(s_{V_x}|_{V_x \cap V_y}) = \varphi(s_{V_y}|_{V_x \cap V_y}) = t|_{V_x \cap V_y}$$

This is sufficient because φ is injective. The sheaf condition implies that $\{s_{V_x}\}$ do glue to yield $s \in \mathcal{F}(U)$ such that $s|_{V_x} = s_{V_x}$. By construction, $\varphi(s)_x = t_x$ for all x . So $\varphi(s) = t$.

3. ($\Leftarrow$) this follows from the first square. ($\Rightarrow$) this is immediate from the functoriality of taking stalks. $\square$

Vocabulary 1.9 (in/sur/bijective morphisms). A morphism of sheaves $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is an injection/bijection/surjection if it is on all stalks.

Remark. Proposition 1 shows that $\mathcal{F} \rightarrow \mathcal{G}$ is injective/bijective/surjective if and only if $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective/bijective/surjective for all $U \in X$ open. However, $\mathcal{F} \rightarrow \mathcal{G}$ being a surjection of sheaves does not necessarily mean $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is a surjection.

Proposition 2. The inclusion $i : \text{Sh}(X) \hookrightarrow \text{Psh}(X)$ has a left adjoint $a : \text{Psh}(X) \rightarrow \text{Sh}(X)$.

Remark. By left adjoint, we mean there exists a natural isomorphism

$$\text{hom}_{\text{Sh}(X)}(a\mathcal{F}, \mathcal{G}) \simeq \text{hom}_{\text{Psh}(X)}(\mathcal{F}, i\mathcal{G})$$

This means we have $\mathcal{F} \rightarrow a\mathcal{F}$ is the universal arrow to a sheaf. Whenever we have a presheaf map $\mathcal{F} \rightarrow i\mathcal{H} = \mathcal{H}$, there exists a unique map $a\mathcal{F} \rightarrow \mathcal{H}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & a\mathcal{F} \\ \downarrow & \swarrow \exists! & \\ \mathcal{H} & & \end{array}$$

Proof. Given $\mathcal{F}$, a presheaf on X , define $a\mathcal{F}$ as

$$\prod_{x \in U} \mathcal{F}_x \supseteq a\mathcal{F}(U) = \left\{ (s_x)_{x \in U} \mid \begin{array}{l} \forall x \in U, \exists \text{ open nbd } x \in W \subseteq U \\ \text{and } t \in \mathcal{F}(W) \text{ s.t. } t_y = s_y, \forall y \in W \end{array} \right\}$$

The restriction map for $U \subseteq V$ is induced by the projection $\prod_{x \in V} \mathcal{F}_x \rightarrow \prod_{x \in U} \mathcal{F}_x$.

Now we check that this is a sheaf. Let $U \subseteq \bigcup_i V_i$ be an open cover. The restriction map is the natural projection map. This will turn $a\mathcal{F}$ into a presheaf. Indeed, let $s_i \in a\mathcal{F}(V_i)$ and $s_j \in a\mathcal{F}(V_j)$. Suppose $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$. Then we have a map $a\mathcal{F}(V_i) \rightarrow a\mathcal{F}(V_i \cap V_j)$ given by $(s_i) \mapsto (s_i)|_{V_i \cap V_j}$. The projection map looks like

$$\prod_{x \in V_i} \mathcal{F}_x \xrightarrow{\text{proj}} \prod_{x \in V_i \cap V_j} \mathcal{F}_x$$

Define $s \in a\mathcal{F}(U)$ such that $s_x = s_{i_x}$ if $x \in V_i$ and $s_x = s_{j_x}$ if $x \in V_j$. By construction, $s|_{V_i} = s_i$ and $s|_{V_j} = s_j$. We also get a natural transformation $\alpha_{\mathcal{F}} : \mathcal{F} \rightarrow ia\mathcal{F}$ given by $\mathcal{F}(U) \rightarrow a\mathcal{F}(U)$, $s \mapsto (s_x)_{x \in U}$.

Now we check that $\alpha_{\mathcal{F}}$ is an isomorphism on stalks. Indeed, the map $\mathcal{F}_x \rightarrow (a\mathcal{F})_x$ is injective because $\mathcal{F}_x \rightarrow \prod (\mathcal{F})_x$ is injective. To show this is surjective, suppose $t \in (a\mathcal{F})_x$. We can find an open neighborhood $U \ni x$ such that t is the equivalence class of (U, s) and $s \in a\mathcal{F}(U)$. By definition, there exists an open neighborhood $x \in V \subseteq U$

¹<https://math.stackexchange.com/questions/1387214/the-induced-map-on-stalks-is-well-defined>

and a section $\sigma \in \mathcal{F}(V)$ such that $s|_V$ is the image of $\sigma \in \prod(\mathcal{F})(V)$. Then (V, σ) defines an element of $\mathcal{F}_x$ mapping to t . In particular, by Proposition 1 (2), we see that if $\mathcal{F}$ is already a sheaf, then $\mathcal{F} \rightarrow ia\mathcal{F}$ is an isomorphism.

If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves, then there exists a unique morphism $a\varphi : a\mathcal{F} \rightarrow a\mathcal{G}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\alpha_{\mathcal{F}}} & ia\mathcal{F} \\ \downarrow \varphi & & \downarrow a\varphi \\ \mathcal{G} & \xrightarrow{\alpha_{\mathcal{G}}} & ia\mathcal{G} \end{array}$$

The map $a\varphi$ is induced by $\prod \mathcal{F}_x \rightarrow \prod \mathcal{G}_x$. The uniqueness follows from Proposition 1 (3). It turned out, $a : \mathbf{Psh}(X) \rightarrow \mathbf{Sh}(X)$ is a functor (because $a \circ \text{id} = \text{id}$ and $(a\varphi) \circ (a\psi) = a(\varphi \circ \psi)$). Further note that given $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ where $\mathcal{G}$ is a sheaf, then $\varphi, a\varphi$ determine each other through $\varphi = \alpha_{\mathcal{G}}^{-1} \circ a\varphi \circ \alpha_{\mathcal{F}}$. Finally, we get that

$$\begin{aligned} \text{hom}_{\mathbf{Psh}(X)}(\mathcal{F}, ia\mathcal{G}) &\xrightarrow{\sim} \text{hom}_{\mathbf{Sh}(X)}(a\mathcal{F}, \mathcal{G}) \\ \mathcal{F} \xrightarrow{\varphi} \mathcal{G} &\longmapsto \alpha_{\mathcal{G}}^{-1} \circ a\varphi \end{aligned}$$

Check that this is a bijection. □

Vocabulary 1.10 (sheafification). The left adjoint a defined in Proposition 2 is called sheafification.

2 Week 2

9/3

Example. Here is an example of sheafification. Let $A \in \mathbf{Ab}$ and $X \in \mathbf{Top}$. Recall that if cA is a constant presheaf on X , then on some open $U \subseteq X$, $A := cA(U) = \{f \in \text{hom}_{\mathbf{Set}}(U, A) | f \text{ is constant}\}$. The sheafification of A is

$$\underline{A} := \{f \in \text{hom}_{\mathbf{Set}}(U, A) | f \text{ is locally constant}\}$$

where A is equipped with discrete topology.

Vocabulary 2.1 (constant sheaf). $\underline{A}$ appeared in the above example is a constant sheaf on X .

Vocabulary 2.2 (direct image). Let $f : X \rightarrow Y$ be a continuous map between topological spaces. We need a way to transport (pre)sheaves on X to (pre)sheaves on Y . For $U \subseteq Y$ open, define

$$f_* : \mathbf{Psh}(X) \rightarrow \mathbf{Psh}(Y) \quad (f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U))$$

as the direct image of $\mathcal{F}$ under f . If $\varphi : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ is a morphism of (pre)sheaves, then the family of maps $\{f_*\varphi_U = \varphi_{f^{-1}(U)} : \mathcal{F}_1(f^{-1}(U)) \rightarrow \mathcal{F}_2(f^{-1}(U))\}$ gives a morphism $f_*\varphi : f_*\mathcal{F}_1 \rightarrow f_*\mathcal{F}_2$. We see that f_* mapped a morphism between (pre)sheaves on X to a morphism between (pre)sheaves on Y . This makes f_* a functor and therefore justifies $f_* : \mathbf{Psh}(X) \rightarrow \mathbf{Psh}(Y)$ and $f_* : \mathbf{Sh}(X) \rightarrow \mathbf{Sh}(Y)$.

Vocabulary 2.3 (inverse image). We also want a map from $\mathbf{Sh}(Y)$ to $\mathbf{Sh}(X)$. Let $U \subseteq X$ be open. First define $f_{\text{pre}}^{-1} : \mathbf{Psh}(Y) \rightarrow \mathbf{Psh}(X)$ by $(f_{\text{pre}}^{-1}\mathcal{G})(U) = \text{colim}_{f(U) \subseteq V} \mathcal{G}(V)$. This can be described as equivalent classes of sections of $\mathcal{G}$ over opens $V \subseteq Y$ which contain $f(U)$. Note that $f(U)$ may not be open in Y . Define

$$f^{-1} : \mathbf{Sh}(Y) \rightarrow \mathbf{Sh}(X) \quad f^{-1} = af_{\text{pre}}^{-1}$$

We can only evaluate $\mathcal{G}$ on open sets. So we approximate $f(U)$ by taking open sets containing it.

Proposition 3. Let $f : X \rightarrow Y$ be a continuous map.

- (a) $f_{\text{pre}}^{-1} : \mathbf{Psh}(Y) \longleftrightarrow \mathbf{Psh}(X) : f_*$ is an adjoint pair.
- (b) $f^{-1} : \mathbf{Sh}(Y) \longleftrightarrow \mathbf{Sh}(X) : f_*$ is an adjoint pair.

Proof sketch. (a) The key point is to see the existence of natural maps

$$\begin{aligned}\varepsilon : f_{\text{pre}}^{-1}f_*\mathcal{F} &\rightarrow \mathcal{F} && \text{"counit of adjunction"} \\ \eta : \mathcal{G} &\rightarrow f_*f_{\text{pre}}^{-1}\mathcal{G} && \text{"unit of adjunction"}\end{aligned}$$

For ε , take $W \subseteq X$ open. Then

$$\begin{array}{ccccc} f_{\text{pre}}^{-1}f_*\mathcal{F}(W) & = & \operatorname{colim}_{f(W) \subseteq U} f_*\mathcal{F}(U) & = & \operatorname{colim}_{f(W) \subseteq U} \mathcal{F}(f^{-1}(U)) \\ & & \searrow \varepsilon_W & & \downarrow \rho_{f^{-1}(U)W} \\ & & & & \mathcal{F}(W) \end{array}$$

$W \subseteq f^{-1}f(W) \subseteq f^{-1}(U)$

For η , take $V \subseteq Y$. We did $f_{\text{pre}}^{-1}f_*$. Now it is time for $f_*f_{\text{pre}}^{-1}$. So

$$\mathcal{G}(V) \xrightarrow{\eta_V} f_*f_{\text{pre}}^{-1}\mathcal{G}(V) \stackrel{2.2}{=} (f_{\text{pre}}^{-1}\mathcal{G})(f^{-1}(V)) \stackrel{2.3}{=} \operatorname{colim}_{ff^{-1}V \subseteq U} \mathcal{G}(U)$$

The map η_V makes sense because $f(f^{-1}(V)) \subseteq V$ makes V an element of the indexing set. Indeed, the indexing condition can be interpreted as $f^{-1}(V) \subseteq f^{-1}(U)$. So the condition is running through all U such that $V \subseteq U$. In particular, $V = U$ is one of them. There are natural transformations going both ways

$$\begin{aligned}\operatorname{hom}_{\mathbf{Psh}(X)}(f_{\text{pre}}^{-1}\mathcal{G}, \mathcal{F}) &\longleftrightarrow \operatorname{hom}_{\mathbf{Psh}(Y)}(\mathcal{G}, f_*\mathcal{F}) \\ (f_{\text{pre}}^{-1}\mathcal{G} \xrightarrow{\varphi} \mathcal{F}) &\longmapsto (\mathcal{G} \xrightarrow{\eta} f_*f_{\text{pre}}^{-1}\mathcal{G} \xrightarrow{f_*\varphi} f_*\mathcal{F}) \\ (f_{\text{pre}}^{-1}\mathcal{G} \xrightarrow{f_{\text{pre}}^{-1}\psi} f_{\text{pre}}^{-1}f_*\mathcal{F} \xrightarrow{\varepsilon} \mathcal{F}) &\longleftarrow (\mathcal{G} \xrightarrow{\psi} f_*\mathcal{F})\end{aligned}$$

They are inverse to each other. Therefore, they are bijections. (b) follows from (a). $\square$

Vocabulary 2.4 (ringed space). Before we do the local version, let us do the general version first:

- (1) A ringed space is a pair $(X, \mathcal{O}_X)$, where X is a space and $\mathcal{O}_X$ is a sheaf of rings on X .
- (2) A morphism of ringed spaces $(f, f^b) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ consists of a map between spaces $f : X \rightarrow Y$ and a homomorphism $f^b : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ between sheaves of rings. For any open $V \subseteq Y$, $f^b(\mathcal{O}_Y(V)) = \mathcal{O}_X(f^{-1}(V))$.

We use $\mathcal{O}_{X,x}$ to denote the stalk of $\mathcal{O}_X$ at $x \in X$.

Example. Some examples of ringed spaces.

1. The sheaf of $\mathbb{R}$ -valued continuous functions $(X, \mathcal{C}_X)$.
2. Let X be a variety over an algebraically closed field K . Take some zariski open $U \subseteq X$.

$$\mathcal{O}_X(U) = \operatorname{hom}_{\text{cont}}(U, \mathbb{A}^1(K))$$

is the sheaf of regular functions on X .

9/5

Remark. From the formula for f_{pre}^{-1} , we see that $(f_{\text{pre}}^{-1}\mathcal{G})_x = \mathcal{G}_{f(x)}$. Similarly, if $\mathcal{G}$ is a sheaf, $(f^{-1}\mathcal{G})_x = \mathcal{G}_{f(x)}$.

Notation. If $i : U \hookrightarrow X$ is an inclusion of an open subspace, we often write $\mathcal{F}|_U = i^{-1}\mathcal{F}$.

Remark. Think of $\mathcal{O}_X$ as a sheaf of functions on X . Recall that $\mathcal{O}_{X,x}$ is the stalk of $\mathcal{O}_X$ at $x \in X$. This is also the germs of functions at x (i.e. equivalence classes of functions near x). If $f(x) \neq 0$, then f is invertible on some small neighborhood of x . Recall that if $\mathfrak{m}_x = \{[f] | f(x) = 0\} \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$, then the complement of $\mathfrak{m}_x$ consists of invertible elements.

Vocabulary 2.5 (locally ringed space). A locally ringed space is a ringed space $(X, \mathcal{O}_X)$ such that every stalk $\mathcal{O}_{X,x}$ is a local ring.

Example. $(X, \mathcal{C}_X)$ is a locally ringed space. Given $x \in X$, let $\mathfrak{m}_x = \{[s] | s(x) = 0\} \subseteq \mathcal{C}_{X,x}$. If $[s] \in \mathcal{C}_{X,x} \setminus \mathfrak{m}_x$, then $s(x) \neq 0$ for some representatives of $[s]$. Due to continuity, there is a neighborhood $U \ni x$ such that for all $y \in U$, $s(y) \neq 0$. In other words, $1/s$ exists on U and $1/(s(y)) = 1/s(y)$. This implies $\mathcal{C}_{X,x} \setminus \mathfrak{m}_x$ consists of invertible elements. So $\mathcal{C}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$. This corresponds to the evaluation map at x

$$\chi : \mathcal{C}_{X,x} \rightarrow \mathbb{R}$$

Note that $\ker(\chi) = \mathfrak{m}_x$ and $\mathcal{C}_{X,x}/\mathfrak{m}_x = \mathbb{R}$.

Vocabulary 2.6 (residue field). Given a locally ringed space $(X, \mathcal{O}_X)$ and $x \in X$. Write $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$. This is called the residue field of x . For any $f \in \mathcal{O}_X(U)$ and any $x \in X$, write $f(x) \in k(x)$ for the image of f under

$$\begin{array}{ccccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_{X,x} & \longrightarrow & \mathcal{O}_{X,x}/\mathfrak{m}_x = k(x) \\ f & \longmapsto & & \longrightarrow & f(x) \end{array}$$

Note that these are ring maps.

Vocabulary 2.7 (maps between locally ringed spaces). To give a map between locally ringed spaces, we first have to provide a map between ringed spaces $(f, f^\flat) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$. Here, $f^\flat : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ has an adjoint map $f^\sharp : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. There are homomorphisms on stalks at $x \in X$, namely, $f_x^\sharp : \mathcal{O}_{Y,f(x)} = (f^{-1}\mathcal{O}_Y)_x \rightarrow \mathcal{O}_{X,x}$. Indeed, for each $U \subseteq Y$ neighborhood of $f(x)$,

$$\mathcal{O}_Y(U) \xrightarrow{f^\flat} (f_*\mathcal{O}_X)(U) = \mathcal{O}_X(f^{-1}(U)) \xrightarrow{\eta} \mathcal{O}_{X,x}$$

Recall that $\eta : \mathcal{F}(U) \rightarrow \mathcal{F}_x$ is the canonical map introduced in Proposition 1. And also note that $f^{-1}(U)$ is a neighborhood of $x \in X$. Finally, this gives a ring homomorphism

$$f_x^\sharp : \mathcal{O}_{Y,f(x)} = \operatorname{colim}_{f(x) \in U \subseteq Y} \mathcal{O}_Y(U) \longrightarrow \mathcal{O}_{X,x}$$

If for all $x \in X$ the homomorphism $f_x^\sharp$ is local, then $(f, f^\flat)$ is a morphism of locally ringed spaces. Recall a homomorphism between local rings $f : A \rightarrow B$ is local if $f(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

3 Week 3

9/8

Recall that $k^n \supseteq X$ is an affine variety over an algebraically closed field. Write $X \simeq \{\mathfrak{m} \subseteq \Gamma(X) \mid \mathfrak{m} \text{ is maximal}\}$, where $\Gamma(X)$ is the coordinate ring of X . The topology on X has closed sets $V(I) = \{x \in X \mid f(x) = 0, \forall f \in I\} = \{\mathfrak{m} \subseteq \Gamma(X) \mid I \subseteq \mathfrak{m}\}$ for some ideal $I \subseteq \Gamma(X)$.

Vocabulary 3.1 (spec). Let A be a ring. Define $\operatorname{spec}(A)$ to be the set of prime ideals of A . Think of A as a ring of functions on $\operatorname{spec}(A)$. So given some function $f \in A$ and $x = \mathfrak{p} \in \operatorname{spec}(A)$, we may define $f(x)$ as the image of f under $A \rightarrow A/\mathfrak{p} \rightarrow \operatorname{frac}(A/\mathfrak{p})$. Here $\operatorname{frac}(-)$ is the fraction field.

Vocabulary 3.2 (Zariski locus of vanishing). Let $M \subseteq A$ be a subset. Define $V(M) = \{\mathfrak{p} \in \operatorname{spec}(A) \mid f(\mathfrak{p}) = 0, \forall f \in M\}$. Recall that $f(\mathfrak{p})$ is the image of f under $A \rightarrow A/\mathfrak{p} \rightarrow \operatorname{frac}(A/\mathfrak{p})$. In other words, $f(\mathfrak{p}) = 0$ implies $(f + \mathfrak{p})/1 = 0$. Conclude that

$$V(M) = \{\mathfrak{p} \in \operatorname{spec}(A) \mid f(\mathfrak{p}) = 0, \forall f \in M\} = \{\mathfrak{p} \in \operatorname{spec}(A) \mid M \subseteq \mathfrak{p}\}$$

Vocabulary 3.3 (Zariski topology on spec). The Zariski topology on $\operatorname{spec}(A)$ is the topology where closed sets are exactly the subsets of $\operatorname{spec}(A)$ of form $V(I)$ for any ideal $I \subseteq A$.

Proposition 4. Let A be a ring. Let I, J be ideals of A . Then

- (0) $I \subseteq J$ implies $V(J) \subseteq V(I)$.
- (1) $V(\langle 0 \rangle) = \operatorname{spec}(A)$.
- (2) Let $\{I_r\}$ be any collection of ideals. Then $V(\bigcup_r I_r) = V(\sum_r I_r) = \bigcap_r V(I_r)$.

$$(3) V(I \cap J) = V(IJ) = V(I) \cup V(J).$$

□

Example. Here are some examples of Zariski spectrum.

1. $\text{spec}(\mathbb{Z}) = \{\langle 0 \rangle\} \cup \{\langle 2 \rangle, \langle 3 \rangle, \langle 5 \rangle, \langle 7 \rangle, \dots\}$. For example, we choose $f = 3 \in \mathbb{Z}$ and $\mathfrak{p} = \langle 3 \rangle \in \text{spec}(\mathbb{Z})$. Its image in $\mathbb{Z} \rightarrow \mathbb{Z}/\langle 3 \rangle = \mathbb{F}_3$ is $f(\langle 3 \rangle) = 0$. If instead we choose $\mathfrak{p} = \langle 2 \rangle$, then $f(\langle 2 \rangle) = [3]_2 \in \mathbb{F}_2$. Recall that the closure of a point $\mathfrak{q} \in \text{spec}(\mathbb{Z})$ is $\overline{\{\mathfrak{q}\}} = \bigcap_{\mathfrak{q} \subseteq S \text{ closed}} S$. But for any prime number P , $\langle P \rangle$ is maximal. So $V(\langle P \rangle) = \{\langle P \rangle\}$ implies every $\langle P \rangle$ is closed. Now the closure becomes $\overline{\{\mathfrak{q}\}} = \bigcap_{\mathfrak{q} \subseteq \mathfrak{p} \in \text{spec}(\mathbb{Z})} \mathfrak{p}$. Note that every prime ideal contains 0. Conclude that

$$\overline{\{\langle 0 \rangle\}} = \{\mathfrak{q} \in \text{spec}(\mathbb{Z}) \mid \langle 0 \rangle \subseteq \mathfrak{q}\} = \text{spec}(\mathbb{Z})$$

2. Let $\mathbb{A}_{\mathbb{C}}^1 = \text{spec}(\mathbb{C}[x]) = \{\langle 0 \rangle\} \cup \{\langle x - a \rangle\}_{a \in \mathbb{C}}$. Here $\langle 0 \rangle$ is prime but not maximal and each $\langle x - a \rangle$ is maximal. Since all such ideals contain 0, we still have $\overline{\{\langle 0 \rangle\}} = \mathbb{A}_{\mathbb{C}}^1 = \mathbb{A}^1(\mathbb{C}) \cup \{\langle 0 \rangle\}$. In general if X is a variety over an algebraically closed field k , then the closed points of $\text{spec}(\Gamma(X))$ are points of $X \subseteq k$ (remember that $\Gamma(X)$ is the coordinate ring). However, it has non-closed points as well: it has one for each irreducible closed subvariety of X .

Vocabulary 3.4 (ideal of subset). Let $Y \subseteq \text{spec}(A)$ be a subset. The ideal of Y is defined as

$$I(Y) = \bigcap_{\mathfrak{p} \in Y} \mathfrak{p} \subseteq A$$

Think of this as functions that vanish at all points of Y . This gives an inclusion reversing assignment

$$\{\text{ideals of } A\} \longleftrightarrow \{\text{subsets of } \text{spec}(A)\}$$

$$I \subseteq A \longmapsto V(I)$$

$$I(Y) \longleftarrow Y \subseteq \text{spec}(A)$$

Proposition 5. Given $I \subseteq A$ an ideal and $Y \subseteq \text{spec}(A)$ a subset,

1. $I(Y) = \text{rad}(I(Y))$.
2. $I(V(I)) = \text{rad}(I)$.
3. The assignment

$$\{\text{radical ideals of } A\} \longleftrightarrow \{\text{closed subsets of } \text{spec}(A)\}$$

is actually an inverse bijection.

Proof. We justify each claim.

1. Let J be an ideal. Then $J = \text{rad}(J)$ if and only if J whenever $f \in A$ satisfies $f^n \in J$, $f \in J$. Note that any intersection of prime ideals will satisfy the latter condition. In particular $I(Y)$.
2. In general, $\text{rad}(J) = \bigcap_{\mathfrak{p} \text{ prime}, \mathfrak{p} \supseteq J} \mathfrak{p}$. So $I(V(I)) = \text{rad}(I)$.
3. Observe that if $Y \subseteq \text{spec}(A)$ is a subset, then $V(I(Y))$ is the smallest closed subset of $\text{spec}(A)$ which contains Y . In other words, $\overline{Y} = V(I(Y))$. In particular, if Y is already closed, then $Y = V(I(Y))$.

□

9/10

Vocabulary 3.5 (principal open). A principal open subset of $\text{spec}(A)$ is an open subset of the form

$$D(f) = \text{spec}(A) \setminus V(f) = \{\mathfrak{p} \in \text{spec}(A) \mid f(\mathfrak{p}) \neq 0\} = \{\mathfrak{p} \in \text{spec}(A) \mid f \notin \mathfrak{p}\}$$

They form a basis for open sets in the Zariski topology. We will later prove this fact. From this definition, we see that ① $D(fg) = D(f) \cap D(g)$ ② $D(0) = \emptyset$ ③ $D(1) = \text{spec}(A)$.

Example. Recall that $\text{spec}(\mathbb{Z}) = \{\langle 0 \rangle\} \cup \{\langle 2 \rangle, \langle 3 \rangle, \langle 5 \rangle, \langle 7 \rangle, \dots\}$ and the "function" f on X is given by $f(p) = [n]_p \in \mathbb{F}_p$. The locus of vanishing is $V(n) = \{p \mid [n]_p = 0\} = \{p \mid p \text{ divides } n\}$. And $D(n) = \{p \mid \gcd(n, p) = 1\} \cup \{\langle 0 \rangle\}$.

Proposition 6. More facts:

1. $\{D(f)\}_{f \in A}$ forms a basis for the zariski topology.
2. $D(g) \subseteq \bigcup_i D(f_i)$ if and only if $g^n \in \sum_i \langle f_i \rangle$ for some n .

Proof. Let us quickly prove these results

1. By Proposition 4, $V(I) = \bigcap_{f \in I} V(f)$. This means every closed set is an intersection of $V(f)$'s. Taking the complement gives every open is a union of $D(f)$'s.
2. $D(g) \subseteq \bigcup_i D(f_i)$ if and only if $\bigcap_i V(f_i) = V(\sum_i \langle f_i \rangle) \subseteq V(g)$. Proposition 5 gives $\text{rad}(\langle g \rangle) \subseteq \text{rad}(\sum_i \langle f_i \rangle)$. It follows that $g^n \in \sum_i \langle f_i \rangle$.

□

Vocabulary 3.6 (spec map). Let $\varphi : A \rightarrow B$ be a homomorphism of rings. If $\mathfrak{q}$ is a prime ideal of B , then $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of A . Therefore we obtain a map

$$\text{spec}(\varphi) : \text{spec}(B) \rightarrow \text{spec}(A) \quad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$$

Proposition 7. Quick facts about the spec map:

1. $\text{spec}(\varphi)^{-1}(V(M)) = V(\varphi(M))$ for any subset $M \subseteq A$. If $f \in A$, then $\text{spec}(\varphi)^{-1}(V(f)) = V(\varphi(f))$.
2. $V(\varphi^{-1}(I)) = \text{spec}(\varphi)(V(I))$ for any ideal $I \subseteq B$.

□

Proposition 8. Let A be a ring.

1. Let $\varphi : A \rightarrow B$ be a surjective homomorphism of rings with kernel $\mathfrak{a}$. Then $\text{spec}(\varphi)$ is a homeomorphism of $\text{spec}(B)$ onto the closed subset $V(\mathfrak{a})$ of $\text{spec}(A)$. Moreover, for any ring R and an ideal $I \subseteq R$, $\varphi : R \rightarrow R/I$ induces a homeomorphism.
2. Let S be a multiplicative subset of A and let $\varphi : A \rightarrow B := A_S$ be the canonical homomorphism. Then $\text{spec}(\varphi)$ is a homeomorphism of $\text{spec}(A_S)$ onto the subspace $\{\mathfrak{p} \in \text{spec}(A) \mid S \cap \mathfrak{p} = \emptyset\} \subseteq \text{spec}(A)$.

Proof. Even though these statements are similar, we are proving them carefully anyway.

1. $\text{spec}(\varphi)$ is injective. Moreover, a prime ideal $\mathfrak{q} \subseteq B$ contains an ideal $\mathfrak{b} \subseteq B$ if and only if $\varphi^{-1}(\mathfrak{b}) \subseteq \varphi^{-1}(\mathfrak{q})$. This shows that $\text{spec}(\varphi)(V(\mathfrak{b})) = V(\varphi^{-1}(\mathfrak{b})) \cap \text{im}(\text{spec}(\varphi))$. Therefore, $\text{spec}(\varphi)$ is a homeomorphism onto its image. Since $\text{spec}(\varphi) : \text{spec}(R/I) \rightarrow V(I)$ is continuous and bijective, we just need to show it is also closed. Let $V(J/I) \subseteq \text{spec}(R/I)$ be closed, where $J/I \subseteq R/I$. J is the corresponding ideal in R . It follows that $(\text{spec}(\varphi))(V(J/I)) = V(J) \subseteq V(I)$.
2. Wording it differently, the localization map $\varphi : A \rightarrow A_S$ induces a homeomorphism

$$\text{spec}(\varphi) : \text{spec}(A_S) \xrightarrow{\sim} \{\mathfrak{p} \in \text{spec}(A) \mid \mathfrak{p} \cap S = \emptyset\} \subseteq \text{spec}(A)$$

From basic properties of localization, $\text{spec}(\varphi)$ is a continuous bijection. In particular, $\text{spec}(\varphi)$ is given by $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{q}) = \mathfrak{q} \cap A$, with inverse $\mathfrak{p}A_S = \mathfrak{p}_S \leftarrow \mathfrak{p}$. For any ideal $J \subseteq A_S$ and primes $\mathfrak{p}, \mathfrak{q}$, $\mathfrak{q} \supseteq J$ implies $\varphi^{-1}(\mathfrak{q}) \supseteq \varphi^{-1}(J)$. If $\varphi^{-1}(\mathfrak{q}) \supseteq \varphi^{-1}(J)$, then for all $x = a/s \in J$, $sx = a/1 \in J$. So $a \in \varphi^{-1}(J) \subseteq \varphi^{-1}(\mathfrak{q})$. This gives $x = a/s \in \varphi^{-1}(\mathfrak{q})A_S = \mathfrak{q}$. Now we have equalities

$$\begin{aligned} \text{spec}(\varphi)(V(J)) &= \{\varphi^{-1}(\mathfrak{q}) \mid \mathfrak{q} \supseteq J, \mathfrak{q} \in \text{spec}(A)\} \\ &= \{\varphi^{-1}(\mathfrak{q}) \mid \varphi^{-1}(\mathfrak{q}) \supseteq \varphi^{-1}(J)\} = \text{im}(\text{spec}(\varphi)) \cap V(\varphi^{-1}(J)) \end{aligned}$$

Thus $\text{spec}(\varphi) : \text{spec}(A_S) \rightarrow \text{im}(\text{spec}(\varphi))$ is also closed. This is a homeomorphism.

□

Now we can define $\mathcal{O}_{\text{spec}(A)}$ on $\text{spec}(A)$. Given a space X with a basis $\mathcal{B}$ for the topology. It suffices to specify a sheaf on $\mathcal{B}$ to specify a sheaf on X . Precisely, given $\mathcal{F} : \mathcal{B}^{\text{op}} \rightarrow \text{Set}/\text{Ab}/\text{RMod}$ and $V \subseteq X$ open, we can define

$$\mathcal{F}'(V) = \lim_{U_i \in \mathcal{B}, U_i \subseteq V} \mathcal{F}(U_i) = \left\{ F \in \prod_{U_i \in \mathcal{B}, U_i \subseteq V} \mathcal{F}(U_i) \mid F_{U_i} = \rho_{U_j U_i}(F_{U_j}), \forall U_i \subseteq U_j \right\}$$

where F is a tuple indexed by U_i 's and $F_{U_i} \in \mathcal{F}(U_i)$. If $V \in \mathcal{B}$, then one of the members in the product is $\mathcal{F}(V)$. So $\iota : \mathcal{F}(V) \rightarrow \mathcal{F}'(V)$ given by $\iota(t) = (\rho_{V U}(t))_{U \subseteq V}$ is an inclusion. On the other hand, we can define $\pi_V : \mathcal{F}'(V) \rightarrow \mathcal{F}(V)$ by $\pi_V : (s_U)_U \mapsto s_V$. Compatibility within $\mathcal{F}$ will give $s_U = \rho_{V U}(s_V)$ for all $U \subseteq V$, which

tells us π_V and ι are inverses. This shows that $\mathcal{F}$ and $\mathcal{F}'$ agree on basis elements. It follows that $\mathcal{F}'$ is a sheaf if and only if $\mathcal{F}$ satisfies sheaf conditions on $\mathcal{B}$: if $U \in \mathcal{B}$ and $U = \bigcup_i U_i$ where $U_i \in \mathcal{B}$, then the following sequence

$$\mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

is exact. Proposition 8 says for any element $f \in A$, $\varphi : A \rightarrow A_f$ induces a homeomorphism between $\text{spec}(A_f)$ and $\{\mathfrak{p} \in \text{spec}(A) \mid f \cap \mathfrak{p} = \emptyset\} = D(f) \subseteq \text{spec}(A)$.

Vocabulary 3.7 (sheaf on spec). Define sheaf $\mathcal{O}_{\text{spec}(A)}$ on $\text{spec}(A)$ by

$$\mathcal{O}_{\text{spec}(A)}(D(f)) = A_f$$

Restriction maps are given as follows: $D(f) \subseteq D(g)$ if and only if $f^n \in A \cdot g$ if and only if $g \in (A_f)^\times$. The first if and only if is given by Proposition 6. The second if and only if is because $f^n \in A \cdot g$ implies $f^n = ag$ for some a . Then $1 = g(a/f^n)$. The universal property of localization gives a unique ring map

$$\rho_{g,f} : A_g = \mathcal{O}_{\text{spec}(A)}(D(g)) \rightarrow \mathcal{O}_{\text{spec}(A)}(D(f)) = A_f$$

Uniqueness tells us $\rho_{h,f} = \rho_{g,f} \circ \rho_{h,g}$ if $D(f) \subseteq D(g) \subseteq D(h)$. This also makes it functorial.

9/12

Theorem 2. $\mathcal{O}_{\text{spec}(A)}$ is a sheaf on $\text{spec}(A)$.

Proof. Suppose $D(f) = \bigcup_i D(f_i)$. There are a few observations:

- ① By Proposition 6, $f^n \in \sum_i \langle f_i \rangle$. So $f^n = \sum_i a_i f_i$ for finitely many $a_i \neq 0$. We can assume $D(f) = \bigcup_i D(f_i)$ is a finite union.
- ② Replacing A by A_f , we have $\mathcal{O}_{\text{spec}(A)}(D(f)) = \mathcal{O}_{\text{spec}(A_f)}(\text{spec}(A_f)) = \mathcal{O}_{\text{spec}(A_f)}(D(1)) = A_f$. This is in the same spirit as $\mathcal{O}_{\text{spec}(A)}(D(1)) = A$. So we may assume $f = 1$ and $f^n = 1 \in \sum_i \langle f_i \rangle$. This implies $\sum_i \langle f_i \rangle = A$.
- ③ Since $p \in \mathfrak{p}$ if and only if $p^n \in \mathfrak{p}$, we have $D(f_i) = D(f_i^n)$. It follows that $\sum_i \langle f_i^n \rangle = A$ for all $n \geq 1$. So for each n , we can find an expression $1 = \sum_i b_i f_i^n$.

If $D(f) = \bigcup_i D(f_i)$, we need to check the following sheaf conditions

- (i) $s \in \mathcal{O}_{\text{spec}(A)}(D(f))$ such that $s|_{D(f_i)} = 0$ for all i , then $s = 0$. Let $s \in \mathcal{O}_{\text{spec}(A)}(\text{spec}(A)) = A$ such that

$$0 = s/1 = s|_{D(f_i)} \in \mathcal{O}_{\text{spec}(A)}(D(f_i)) = A_{f_i}$$

for all i . In other words, $s = a/f_i^r = 0 \in A_{f_i}$ for some r . This implies $f_i^r s = 0 \in A$. By ①, there are only finitely many i . We can choose some $N \gg 0$ so that $f_i^N s = 0$ for all i . By ③, $1 = \sum_i b_i f_i^N$. So $s = \sum_i (b_i f_i^N s) = 0$.

- (ii) Given $s_i \in \mathcal{O}_{\text{spec}(A)}(D(f_i))$ so that $s_i|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)}$, then there is $s \in \mathcal{O}_{\text{spec}(A)}(D(f))$ such that $s|_{D(f_i)} = s_i$ for all i . Suppose we are given $s_i \in A_{f_i}$. Write $s_i = a_i/f_i^N$, where N is independent of i , just like what we did earlier. Also suppose $s_i|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)}$. This means $a_i/f_i^N = a_j/f_j^N$ in $A_{f_i f_j}$. So

$$0 = (f_i f_j)^r \left(\frac{a_i}{f_i^N} - \frac{a_j}{f_j^N} \right) = \frac{(f_i f_j)^r}{(f_i f_j)^N} (f_j^N a_i - f_i^N a_j)$$

Here, we can choose r, N independent of i with $r > N$. Let $m = r - N$. Write $f_i^m f_j^m (f_j^N a_i - f_i^N a_j) = 0$ for all i, j . This gives us a formula $f_j^{m+N} f_i^m a_i = f_i^{m+N} f_j^m a_j$. By observation ③, we can write $1 = \sum_i b_i f_i^{m+N}$ for some b_i . Now define $s := \sum_j b_j f_j^m a_j \in A$. We claim that in A_{f_i} , $s = s_i = a_i/f_i^N$. Indeed,

$$f_i^{N+m} s = f_i^{N+m} \sum_j b_j f_j^m a_j = \sum_j b_j f_j^m f_i^{N+m} a_j \stackrel{\text{formula}}{=} \left(\sum_j b_j f_j^{N+m} \right) f_i^m a_i = f_i^m a_i$$

In the localization $A_{f_i} = \mathcal{O}_{\text{spec}(A)}(D(f_i))$, $s = f_i^m a_i / f_i^{N+m} = a_i / f_i^N = s_i$. □

Remark. Suppose $X = \operatorname{spec}(A)$. Take $x = \mathfrak{p} \in \operatorname{spec}(A)$. The stalk at x is

$$\mathcal{O}_{X,x} = \operatorname{colim}_{x \in D(f)} \mathcal{O}_X(D(f)) = \operatorname{colim}_{f \notin \mathfrak{p}} A_f = \operatorname{colim}_{f \notin \mathfrak{p}} A[1/f] = A_{\mathfrak{p}} = \{a/b \mid b \notin \mathfrak{p}\}$$

Recall that taking a colimit over A_f means we are considering the disjoint union of all such A_f . We identify elements that agree in some further localization A_{fg} . The last equality becomes clear after writing things in fractions.

Remark. Localizing at a prime ideal produces a local ring. It follows that $\mathcal{O}_{X,x}$ is local for all x . Therefore, $(\operatorname{spec}(A), \mathcal{O}_{\operatorname{spec}(A)})$ is a locally ringed space. Recall that every ring map $\varphi : A \rightarrow B$ has a dual map $f : \operatorname{spec}(B) \rightarrow \operatorname{spec}(A)$ given by $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. This induces a direct map and hence $f^b : \mathcal{O}_{\operatorname{spec}(A)} \rightarrow f_* \mathcal{O}_{\operatorname{spec}(B)}$. For $z \in A$,

$$\begin{aligned} \mathcal{O}_{\operatorname{spec}(A)}(D(z)) &= A[1/z] \xrightarrow{f^b_{D(z)}} \mathcal{O}_{\operatorname{spec}(B)}(f^{-1}D(z)) \stackrel{(*)}{=} \mathcal{O}_{\operatorname{spec}(B)}(D(\varphi(z))) = B[1/\varphi(z)] \\ a/z^r &\longmapsto \varphi(a)/\varphi(z)^r \end{aligned} \quad (\ominus)$$

The $(*)$ is concluded by seeing $\mathfrak{q} \in D(\varphi(z))$ if and only if $\varphi^{-1}(\mathfrak{q}) \in D(z)$. Taking colimit over $z \notin \mathfrak{q}$ like in the previous remark, we observe that f^b induces a local ring homomorphism $\varphi : A_{\mathfrak{p}} \rightarrow B_{\varphi^{-1}(\mathfrak{p})}$.

4 Week 4

9/15

Vocabulary 4.1 (scheme). A scheme is a locally ringed space $(X, \mathcal{O}_X)$ such that there exist locally ringed spaces $(U_i, \mathcal{O}_{U_i})$ with properties that ① $X = \bigcup_i U_i$ is an open cover, ② $\mathcal{O}_X|_{U_i} = \mathcal{O}_{U_i}$, and ③ $(U_i, \mathcal{O}_{U_i}) \simeq (\operatorname{spec}(A_i), \mathcal{O}_{\operatorname{spec}(A_i)})$ as a locally ringed space for some ring A_i .

Vocabulary 4.2 (morphism of schemes). A morphism of scheme is a morphism of locally ringed spaces.

Vocabulary 4.3 (affine scheme). An affine scheme is a scheme of the form $(\operatorname{spec}(A), \mathcal{O}_{\operatorname{spec}(A)})$. Write **Sch** as the category of schemes. Write **Aff** $\subseteq$ **Sch** as the full subcategory of affine schemes.

Vocabulary 4.4 (section). Given a ringed space $(Z, \mathcal{O}_Z)$ and some open subset $W \subseteq Z$. We call the action of evaluating $\mathcal{O}_Z(W)$ "taking a section". Denoted as $\Gamma(W, \mathcal{O}_Z)$.

Example. Let us see some examples of schemes.

1. Let R be a ring. $\mathbb{A}_R^n := \operatorname{spec}(R[x_1, \dots, x_n])$. Note that any $\vec{r} \in R^n$ determines a closed point in $\mathbb{A}_R^n$ by forming a maximal ideal like $\langle \vec{x} - \vec{r} \rangle \in \mathbb{A}_R^n$. We use $\vec{r}$ to represent this point. However, *not* all closed points of $\mathbb{A}_R^n$ are obtained via maximal ideals like this *unless* $R = \bar{k}$ is an algebraically closed field. For example, let $\mathbb{A}_{\mathbb{R}}^1 = \operatorname{spec}(\mathbb{R}[x])$. The maximal ideal $\alpha = \langle x^2 + 1 \rangle \in \operatorname{spec}(\mathbb{R}[x])$ is a closed point. Note that this ideal need not to be of the form $\langle \vec{x} - \vec{r} \rangle$ because $\mathbb{R}$ is not algebraically closed. Now compute the residue field

$$k(\alpha) = \mathcal{O}_{\operatorname{spec}(\mathbb{R}[x]), \alpha} / \mathfrak{m}_{\alpha} \simeq \mathbb{R}[x]_{\alpha} / \alpha \mathbb{R}[x]_{\alpha} \simeq {}^2\operatorname{frac}(\mathbb{R}[x]/\alpha) = \mathbb{R}[x]/\alpha \simeq \mathbb{C}$$

2. Take an open subset of $\mathbb{A}_R^1 = \operatorname{spec}(R[x]) \supseteq \mathbb{A}_R^1 \setminus \{0\} = D(x)$. Later, we will see $D(x) \simeq \operatorname{spec}(R[x, x^{-1}])$.
3. Take an open subset $X = \mathbb{A}_R^2 \setminus \{(0, 0)\} \subseteq \mathbb{A}_R^2 = \operatorname{spec}(R[x, y])$. We claim that X is a non-affine scheme. If $(X, \mathcal{O}_X)$ is a scheme, then take any open $U \subseteq X$, there exists a scheme structure on U , namely $(U, \mathcal{O}_X|_U)$. This fact is later justified in Week 5. Get an open cover $X = D(x) \cup D(y)$. Think of this as $\mathbb{R}^2$ minus two axes. By the sheaf construction on spec ,

$$\Gamma(D(x), \mathcal{O}_X) = R[x, y, x^{-1}]; \quad \Gamma(D(y), \mathcal{O}_X) = R[x, y, y^{-1}]; \quad \Gamma(D(x) \cap D(y), \mathcal{O}_X) = R[x, y, x^{-1}, y^{-1}]$$

The global section $\Gamma(X, \mathcal{O}_X)$ can be computed using the sheaf condition. Indeed, suppose $s_1 \in \Gamma(D(x), \mathcal{O}_X)$ and $s_2 \in \Gamma(D(y), \mathcal{O}_X)$. If $s_1|_{D(x) \cap D(y)} = s_2|_{D(x) \cap D(y)}$, then there is an $s \in \Gamma(X, \mathcal{O}_X)$ such that $s|_{D(x)} = s_1$ and $s|_{D(y)} = s_2$. Note that s_1 and s_2 agree on the intersection means the maps

$$\Gamma(D(x), \mathcal{O}_X) \rightarrow \Gamma(D(x) \cap D(y), \mathcal{O}_X) \quad \Gamma(D(y), \mathcal{O}_X) \rightarrow \Gamma(D(x) \cap D(y), \mathcal{O}_X)$$

have to agree. But they only agree on $R[x, y, y^{-1}] \cap R[x, y, x^{-1}] = R[x, y]$. This is where every s comes from as well. So $\Gamma(X, \mathcal{O}_X) = R[x, y]$. Later we will show **Aff** $\simeq$ **Ring**^{op} given by Γ and spec . This implies X cannot be affine. Otherwise $X \simeq \mathbb{A}_R^2$.

²See <https://math.stackexchange.com/questions/4357364/operatornamefracr-p-cong-r-p-pr-p-rq=1>

Remark. $\{0, 0\} \hookrightarrow \mathbb{A}_R^2$ is a codimension 2 subscheme. Later we will see that for "good" schemes, we always have the property that if $z \subseteq X$ is a closed subscheme such that $\text{codim}_Z(X) \geq 2$, then $\Gamma(X \setminus Z, \mathcal{O}_X) \simeq \Gamma(X, \mathcal{O}_X)$.

Proposition 9. The functor $\text{spec} : \text{Ring}^{\text{op}} \rightarrow \text{Aff}$ is an equivalence of categories.

Proof. We need to check the spec map $\rho : \text{hom}_{\text{Ring}}(B, A) \rightarrow \text{hom}_{\text{Aff}}(\text{spec}(A), \text{spec}(B))$ given by $(f : B \rightarrow A) \mapsto ((\rho_f, \rho_f^b) : (\text{spec}(A), \mathcal{O}_{\text{spec}(A)}) \rightarrow (\text{spec}(B), \mathcal{O}_{\text{spec}(B)}))$ is an isomorphism. In other words, we want to show the map $\Gamma : \text{Aff} \rightarrow \text{Ring}^{\text{op}}$ given by $(X, \mathcal{O}_X) \mapsto \Gamma(X, \mathcal{O}_X)$ is the inverse map of spec.

$$\begin{aligned} \text{hom}_{\text{Ring}}(B, A) &\xrightarrow{\rho} \text{hom}_{\text{Aff}}(\text{spec}(A), \text{spec}(B)) \xrightarrow{\Gamma} \text{hom}_{\text{Ring}}(B, A) \\ f : B \rightarrow A &\longmapsto (\rho_f, \rho_f^b) \longmapsto \Gamma(\rho_f^b) : \Gamma(\text{spec}(B), \mathcal{O}_{\text{spec}(B)}) \rightarrow \Gamma(\text{spec}(A), \mathcal{O}_{\text{spec}(A)}) \end{aligned}$$

This shows injectivity because from **(O)**, we know for any $x \in B$, $\rho_f^b : \mathcal{O}_{\text{spec}(B)}(D(x)) \rightarrow \mathcal{O}_{\text{spec}(A)}(D(f(x)))$ turns out to be the map $B[1/x] \rightarrow A[1/f(x)]$ given by $b/x^r \mapsto f(b)/f(x)^r$. If we choose $x = 1$, then

$$\rho_f^b|_{x=1} : \mathcal{O}_{\text{spec}(B)}(D(1)) = B \rightarrow \mathcal{O}_{\text{spec}(A)}(D(f(1))) = A_{f(1)} = A \quad b/1 \mapsto f(b)/1$$

Note that $\mathcal{O}_{\text{spec}(B)}(D(1)) = \Gamma(\text{spec}(B), \mathcal{O}_{\text{spec}(B)})$. This proves $\Gamma(\rho_f^b) = \rho_f^b|_{x=1} = f$. This shows that $\Gamma \circ \rho$ is the identity map. Therefore, ρ is injective. It remains to show surjectivity. Let $X = \text{spec}(A)$ and $Y = \text{spec}(B)$. For any $(\psi, \psi^b) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$, we write down the induced map $\chi := \Gamma(\psi^b) : B = \Gamma(Y, \mathcal{O}_Y) \rightarrow A = \Gamma(X, \mathcal{O}_X)$. To show surjectivity, our strategy is to first pick any affine morphism like (ψ, ψ^b) . And then show there is a ring map that will get mapped to (ψ, ψ^b) under ρ . Our proposed ring map is χ . So in other words, we are trying to show $(\rho_\chi, \rho_\chi^b) = (\psi, \psi^b)$. For any $x \in \text{spec}(A)$, let $\mathfrak{q}_x \subseteq A$ be the corresponding prime ideal. Similarly, for the corresponding element in the image $\psi(x) \in \text{spec}(B)$, let $\mathfrak{p}_{\psi(x)} \subseteq B$. Consider

$$\begin{array}{ccc} B & \xrightarrow{\chi} & A \\ \downarrow \text{loc} & & \downarrow \text{loc} \\ B_{\mathfrak{p}_{\psi(x)}} & \xrightarrow{f_x^\#} & A_{\mathfrak{q}_x} \end{array}$$

Recall that $f_x^\#$ is a local homomorphism. So³ $\mathfrak{p}_{\psi(x)} = (f_x^\#)^{-1}(\mathfrak{q}_x)$. Pulling it back, we get $\mathfrak{p}_{\psi(x)} = \chi^{-1}(\mathfrak{q}_x)$. Observe that $\rho_\chi(x) = \text{spec}(\chi)(\mathfrak{q}_x) = \chi^{-1}(\mathfrak{q}_x)$. This implies $\rho_\chi(x) = \psi(x)$ and hence $\rho_\chi = \psi$. We see that the underlying ring maps are the same. This proves surjectivity. $\square$

Remark. Proposition 9 justified the preceding examples.

9/17

Proposition 9 suggests that spec and Γ are inverses. Therefore, they are bijections in the category of affine schemes. But what happens when we ascend to Sch?

Theorem 3 (gluing schemes). Given schemes $(U_i, \mathcal{O}_{U_i})$ and open subschemes $U_{ij} \subseteq U_i$ with morphisms $\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$ for all i, j such that $U_{ii} = U_i$ and $\varphi_{ii} = \text{id}_{U_i}$. Moreover, $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ on $U_{ij} \cap U_{ki}$. Then there exists a scheme $(X, \mathcal{O}_X)$ such that X contains each U_i as open subschemes. $\square$

Theorem 4. Let X be a scheme and $Y = \text{spec}(A)$. Then the map

$$\Gamma : \text{hom}_{\text{Sch}}(X, Y) \longrightarrow \text{hom}_{\text{Ring}}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \quad f \longmapsto f^\# : \mathcal{O}_Y(Y) \rightarrow f_*\mathcal{O}_X(Y) = \mathcal{O}_X(X)$$

is a bijection.

Proof. Observe the naturality of Γ . For $g : Z \rightarrow X$,

$$\begin{array}{ccc} \text{hom}(X, Y) & \xrightarrow{\Gamma} & \text{hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \downarrow (-) \circ g & & \downarrow g^\#(X) \circ (-) \\ \text{hom}(Z, Y) & \xrightarrow{\Gamma} & \text{hom}(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \end{array}$$

³<https://stacks.math.columbia.edu/tag/07BH>

commutes. Now take $Z = \bigcup_i U_i \supseteq X$ to be an affine open cover of X . This leads to a commuting diagram

$$\begin{array}{ccc} \mathrm{hom}(X, Y) & \xrightarrow{\Gamma} & \mathrm{hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \downarrow \pi^* & & \downarrow \Pi_i \rho_{XU_i} \\ \prod_i \mathrm{hom}(U_i, Y) & \xrightarrow{\prod_i \Gamma_i} & \prod_i \mathrm{hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(U_i)) \end{array}$$

If a morphism is zero on every covering open set, then it must be zero on the whole set. This implies π^* is injective. Now take $f, g : X \rightarrow Y$. Suppose $\Gamma(f) = \Gamma(g) : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$. We apply ρ_{XU_i} to $\Gamma(f) = \Gamma(g)$ so that they become the same map $\Gamma_i(f|_{U_i}) = \Gamma_i(g|_{U_i}) : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(U_i)$. This is the same as taking the section of $f|_{U_i}$ or $g|_{U_i}$. Since $\prod_i \Gamma_i$ is a bijection due to Proposition 9, $f|_{U_i} = g|_{U_i}$. It follows that $f = g$. So Γ is injective. Let $\varphi \in \mathrm{hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$. Since each Γ_i is surjective on U_i by Proposition 9, there exists some $h_i \in \mathrm{hom}(U_i, Y)$ such that $\rho_{XU_i} \circ \varphi = \Gamma_i \circ h_i$. We need to check h_i agree on overlaps. Take any affine open $V \subseteq U_i \cap U_j$. Note that

$$\begin{aligned} h_i|_V : V \hookrightarrow U_i \rightarrow Y &\iff \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(U_i) \rightarrow \mathcal{O}_X(V) \\ h_j|_V : V \hookrightarrow U_j \rightarrow Y &\iff \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(U_j) \rightarrow \mathcal{O}_X(V) \end{aligned}$$

They are equal because $\rho_{U_i V} \circ \rho_{XU_i} = \rho_{U_j V} \circ \rho_{XU_j}$. Going back to the scheme map, we have $h_i|_V = h_j|_V$. They glue to some h such that $h|_{U_i} = h_i$ and $h|_{U_j} = h_j$ by Theorem 3. By the injectivity of $\prod_i \rho_{XU_i}$, $\Gamma(h) = \varphi$. $\square$

Example. We try to actually glue something.

1. Glue two copies $\mathbb{A}_R^1$ along $\mathbb{A}_R^1 \setminus \{0\}$ via a gluing isomorphism $\mathrm{id} : \mathbb{A}_R^1 \setminus \{0\} \rightarrow \mathbb{A}_R^1 \setminus \{0\}$. In Theorem 3 terms, we have $U_1 = \mathbb{A}_R^1 = \mathrm{spec}(R[x]) = U_2$. And $U_{12} = U_{21} = \mathbb{A}_R^1 \setminus \{0\} = D(x) \subseteq U_1 = U_2$ by an earlier example. So the gluing isomorphism is just the induced identity map from $R[x, x^{-1}]$ to itself. Write the glued scheme $X = \mathbb{A}_R^1 \cup_{D(x)} \mathbb{A}_R^1$. This identified every point on the affine lines except for $\{0\}$. So we get an affine line with doubled origins 0_1 and 0_2 . Note that $X \not\cong \mathrm{spec}(R[x])$ because both 0_1 and 0_2 will get mapped to $\langle x \rangle$. Thus X is not affine.
2. We keep the old the setup from (1.). But this time, we define

$$\mathbb{P}_R^1 := \mathbb{A}_R^1 \cup_{\mathbb{A}_R^1 \setminus \{0\}} \mathbb{A}_R^1 : \begin{cases} \varphi_{12} : U_{12} = \mathrm{spec}(R[x, x^{-1}]) \longrightarrow U_{21} = \mathrm{spec}(R[y, y^{-1}]) \\ \Gamma(\varphi_{12}) : R[y, y^{-1}] \longrightarrow R[x, x^{-1}] & y \longmapsto 1/x \end{cases}$$

Suppose $k = \bar{k}$ is an algebraically closed field. Closed points in U_1 are maximal ideals like $\langle x - a \rangle$, where $a \in k$. Our gluing isomorphism does not include the $a = 0$ case. We identify $\langle x \rangle$ as 0 and $\langle y \rangle$ as ∞ . So closed points of $\mathbb{P}_k^1$ are $k \cup \{\infty\}$. We can compute its global section using the sheaf exact sequence

$$0 \longrightarrow \Gamma(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}_k^1}) \longrightarrow \Gamma(U_1, \mathcal{O}_{\mathbb{P}_k^1}) \oplus \Gamma(U_2, \mathcal{O}_{\mathbb{P}_k^1}) \xrightarrow{\chi} \Gamma(U_{12} = U_{21}, \mathcal{O}_{\mathbb{P}_k^1})$$

Calculate $\Gamma(U_1, \mathcal{O}_{\mathbb{P}_k^1}) = R[x]$, $\Gamma(U_2, \mathcal{O}_{\mathbb{P}_k^1}) = R[y]$, $\Gamma(U_{12}, \mathcal{O}_{\mathbb{P}_k^1}) = R[x, x^{-1}]$, and $\Gamma(U_{21}, \mathcal{O}_{\mathbb{P}_k^1}) = R[y, y^{-1}]$. Note that $\chi : R[x] \oplus R[y] \rightarrow R[x, x^{-1}]$ is given by $(f(x), g(y)) \mapsto (f(x) - g(1/x))$. By exactness, $\Gamma(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}_k^1}) = \ker(\chi)$.

But x and x^{-1} only cancel out if and only if we have constant polynomials to begin with. So $\Gamma(\mathbb{P}_k^1, \mathcal{O}_{\mathbb{P}_k^1}) = k$.

Vocabulary 4.5 (irreducible space). A nonempty space is irreducible if it is not a union of two proper closed subsets. A maximal irreducible subset of a space X is called an irreducible component of X .

Proposition 10. Let X be a nonempty space. The followings are the same.

1. X is irreducible.
2. If $U, V \subseteq X$ are nonempty open sets, then $U \cap V \neq \emptyset$.
3. Every nonempty open $U \subseteq X$ is dense.
4. Every nonempty open $U \subseteq X$ is connected.
5. Every nonempty open $U \subseteq X$ is irreducible.

Proof. (1.) $\iff$ (2.) This is immediate by taking complements. (2.) $\iff$ (3.) $U \subseteq X$ is dense if and only if U meets every open set nontrivially. (3.) $\iff$ (4.) If $U \cap V = \emptyset$, then $U \cup V$ is open but not connected. On the other hand, if $W \subseteq X$ is open and disconnected, then we can write W as a disjoint union of opens. (5.) $\implies$ (1.) Clear. (3.) $\implies$ (5.) Every nonempty open set $U \subseteq V$ is dense in V because it is dense in X . So V is irreducible. $\square$

9/19

Vocabulary 4.6 (irreducible scheme). A scheme is irreducible if its underlying topological space is.

Vocabulary 4.7 (specialization). Let X be a topological space. Let $x, y \in X$. y is a specialization of x if $y \in \overline{\{x\}}$.

Vocabulary 4.8 (generic point). Let X be a topological space. Let $x \in X$. x is a generic point of X if $X = \overline{\{x\}}$.

Proposition 11. Let X be a scheme.

1. If $X = \text{spec}(A)$, then the correspondence

$$X \xrightarrow{\sim} \left\{ \begin{array}{l} \text{closed irreducible} \\ \text{subsets of } X \end{array} \right\}$$

$$x \in X \longmapsto \overline{\{x\}} = V(\mathfrak{p}_x)$$

is a bijection. In particular, x (corresponding to $\mathfrak{p}_x \subseteq A$) is a generic point of an irreducible component of X if and only if $\mathfrak{p}_x$ is a minimal prime.

2. The correspondence in (1.) holds for any scheme X .

Proof. (2.) essentially follows from (1.):

1. We have $I(\{x\}) = \mathfrak{p}_x$. By Proposition 5, $\overline{\{x\}} = V(I(\{x\})) = V(\mathfrak{p}_x)$ is closed and irreducible. Now it is time for the other direction of the correspondence. If $Y \subseteq \text{spec}(A)$ is irreducible, then we must show $I(Y)$ is prime. Let $f, g \in A$ and suppose $fg \in I(Y)$. This gives $Y \subseteq V(fg) = V(f) \cup V(g)$. Since Y is irreducible, $Y \subseteq V(f)$ or $Y \subseteq V(g)$. So either $f \in I(Y)$ or $g \in I(Y)$. Again, this then follows loosely from Proposition 5 because prime ideals are radical. Bijection gives uniqueness. So each closed irreducible piece has one generic point.
2. We need to see every closed irreducible $Z \subseteq X$ has a unique generic point as well. Let $U \subseteq X$ be some affine open set such that $Z \cap U \neq \emptyset$. Then in the subspace topology, $Z \cap U \subseteq U$ is closed in U and $Z \cap U \subseteq Z$ is open in Z . It is also irreducible. By (1.), it has a unique generic point. By Proposition 10, $Z \cap U$ is dense in Z . So $\overline{Z \cap U} = Z$. It then follows that a generic point of $Z \cap U$ is also a generic point of Z . Since any generic point $\alpha \in Z$ must be in $Z \cap U$ (otherwise $\overline{\alpha} \subseteq Z \setminus Z \cap U \subsetneq Z$), it is unique by (1.).

□

Example. Let $k = \bar{k}$ be an algebraically closed field. The scheme $\mathbb{A}_k^2$ has 3 kinds of point.

1. Closed points of $\mathbb{A}_k^2$ are maximal ideals in $k[x, y] = \Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}_k^2})$. Since algebraically closed, every maximal ideal looks like $\langle x - a, y - b \rangle$ for $(a, b) \in k^2$.
2. Nonclosed points of $\mathbb{A}_k^2$:
 - (a) Every irreducible polynomial $f \in k[x, y]$ gives a prime $\langle f \rangle$. The closure of $\{\langle f \rangle\}$ consists of $\{(a, b) | f(a, b) = 0\}$ and $\langle f \rangle$. Note that $\overline{\{\langle f \rangle\}} = V(\langle f \rangle) = \{\text{primes containing } f\}$. Since $k[x, y]$ is a UFD, $\langle f \rangle$ is prime. By Week 6 and Week 8 of AP Geometry, $\dim(k[x, y]) = 2$ and $\text{ht}(\langle f \rangle) = 1$. Suppose any prime $\mathfrak{p} \supseteq \langle f \rangle$, then $\text{ht}(\mathfrak{p}) = 2$. This implies that $\mathfrak{p}$ is maximal. So $\overline{\{\langle f \rangle\}} = \{\text{maximal ideals containing } f\} \cup \{\langle f \rangle\}$.
 - (b) The zero ideal $\langle 0 \rangle$ is a generic point of $\mathbb{A}_k^2$. In other word, $\overline{\{0\}} = \mathbb{A}_k^2$.

Example. We take this fact for granted: $\text{spec}(A/I)$ is a closed subscheme of $\text{spec}(A)$. Here $\text{spec}(k[x, y]/\langle xy \rangle) \subseteq \text{spec}(k[x, y]) = \mathbb{A}_k^2$. In fact, we have a homeomorphism $\text{spec}(k[x, y]/\langle xy \rangle) \rightarrow V(\langle xy \rangle)$. But by Proposition 4, $V(\langle xy \rangle) = V(\langle x \rangle) \cup V(\langle y \rangle)$ corresponds to the union of two axes. Each axis is irreducible (will show later). Images of x and y are nonzero in the coordinate ring $k[x, y]/\langle xy \rangle$, But $\overline{xy} = 0$. This gives rise to zero divisors.

Vocabulary 4.9 (connected scheme). A scheme is connected if its underlying topological space is.

Vocabulary 4.10 (quasi-compact scheme). A scheme is quasi-compact if its underlying topological space is.

Proposition 12. $\text{spec}(A)$ is connected if and only if A has no nontrivial idempotents.

Proof. Suppose A contains only nontrivial idempotents. Let e be some idempotent element. We have $\langle e \rangle + \langle 1 - e \rangle = \langle 1 \rangle$ and $\langle e \rangle \cdot \langle 1 - e \rangle = 0$. This then implies $V(\langle e \rangle) = D(\langle 1 - e \rangle)$ by computing complements. So $V(\langle e \rangle)$ and $V(\langle 1 - e \rangle)$ are both open and closed. Observe that $V(\langle e \rangle) \cup V(\langle 1 - e \rangle) = V(\langle e \rangle \cdot \langle 1 - e \rangle) = V(0) = X$ and $V(\langle e \rangle) \cap V(\langle 1 - e \rangle) = V(\langle e \rangle + \langle 1 - e \rangle) = V(1) = \emptyset$. It follows that X is disconnected.

Conversely, let I, J be ideals of A such that $V(I) \cup V(J) = V(I \cap J) = X$ and $V(I) \cap V(J) = \emptyset$. It follows that $X = \{\mathfrak{p} | I \cap J \subseteq \mathfrak{p}\}$. This means $I \cap J \subseteq \bigcap_i \mathfrak{p}_i = \text{nilrad}(A)$. On the other hand, $\emptyset = \{\mathfrak{p} | I + J \subseteq \mathfrak{p}\}$ implies that $1 \in I + J$. Choose $x \in I$ and $y \in J$ with $x + y = 1$. We have that $x^2 - xy = x$. So $x^2 - x \in I \cap J$ is nilpotent. x becomes idempotent in $A/\text{nilrad}(A)$. By Theorem 1A, we get a nontrivial idempotent $e \in A$. □

Example. Some examples (statements) regarding compactness.

1. Affine schemes are quasi-compact.
2. $\coprod_{\mathbb{N}} \text{spec}(k)$ is not quasi-compact and therefore, not affine. For finite cases, $\text{spec}(A \times B) \simeq \text{spec}(A) \coprod \text{spec}(B)$. Inductively, $\text{spec}(\prod_{i=1}^n \text{spec}(A_i)) \simeq \prod_{i=1}^n \text{spec}(A_i)$. But $\text{spec}(\prod_{\mathbb{N}} \text{spec}(A_i)) \not\simeq \prod_{\mathbb{N}} \text{spec}(A_i)$.

Vocabulary 4.11 (open immersion). An open immersion $j : Y \rightarrow X$ between two schemes is a morphism of schemes such that

1. the map on underlying topological spaces is a homeomorphism from Y onto an open subset of $U \subseteq X$.
2. the map $\mathcal{O}_X \rightarrow j_*\mathcal{O}_Y$ induces an isomorphism of sheaves on U . In other words, $\mathcal{O}_X|_U \simeq j_*\mathcal{O}_Y$.

5 Week 5

9/22

Proposition 13. Let $U \subseteq X$ be open. $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof. We first need to show every stalk of $\mathcal{O}_X|_U$ is a local ring. Because it is a restriction from $\mathcal{O}_X$, stalks of $\mathcal{O}_X|_U$ are stalks of $\mathcal{O}_X$. Since $\mathcal{O}_{X,x}$ is a local ring at any $x \in X$, $((\mathcal{O}_X)_U)_x$ is a local ring. Now we need to show it is locally affine. Given any point $x \in U$, we can find some open affine scheme $x \in \text{spec}(R) \subseteq X$. Since the intersection $x \in \text{spec}(R) \cap U \subseteq \text{spec}(R)$ is open in $\text{spec}(R)$ and principal opens form a basis, there is an open $D(f) = \text{spec}(R_f) \subseteq \text{spec}(R) \cap U$ such that $x \in D(f)$. In particular, $D(f) \subseteq U$ is open. This shows $(U, \mathcal{O}_X|_U)$ has an open cover consisting of affine schemes. $\square$

Remark. It follows that $(U, \mathcal{O}_X|_U) \rightarrow (X, \mathcal{O}_X)$ is an open immersion. Moreover, $(U, \mathcal{O}_X|_U)$ is the unique scheme structure on U so that $U \subseteq X$ is a map of schemes.

Q. But what about closed immersions?

A. They are more subtle. Consider an ideal $J \subseteq A$. We have a homeomorphism $\text{spec}(A/J) \simeq V(J)$ by Proposition 8. We also know that $V(J) \subseteq \text{spec}(A)$ is closed subset. But now we want to view $\text{spec}(A/J) \subseteq \text{spec}(A)$ as a closed subscheme. The issue arises as there could be other ideals that can give $V(J)$. In other words, there exists $J \neq J'$ such that $V(J) = V(J') \subseteq \text{spec}(A)$. This will lead to two distinct scheme structures $(\text{spec}(A/J), \mathcal{O}_{\text{spec}(A/J)})$ and $(\text{spec}(A/J'), \mathcal{O}_{\text{spec}(A/J')})$ on $V(J) \subseteq \text{spec}(A)$. Later we will show that we do not have to worry about this.

Vocabulary 5.1 (closed subscheme). A closed subscheme of $(X, \mathcal{O}_X)$ is a closed subset $Z \subseteq X$ with an inclusion $i : Z \hookrightarrow X$. There is also a sheaf $\mathcal{O}_Z$ on Z such that $(Z, \mathcal{O}_Z)$ is a scheme and the map $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$ induces an isomorphism $\mathcal{O}_X/J \simeq i_*\mathcal{O}_Z$ for some sheaf of ideals $J \subseteq \mathcal{O}_X$. As a reminder, for some subset $U \subseteq X$ the evaluation $J(U) \subseteq \mathcal{O}_X(U)$ is an ideal of the ring $\mathcal{O}_X(U)$. And $\mathcal{O}_X(U)/J(U)$ is a quotient ring.

Vocabulary 5.2 (closed immersion). A morphism $i : Z \rightarrow X$ is a closed immersion if the underlying map of spaces is a homeomorphism from Z to a closed subspace of X and $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$ is a surjection of sheaves.

Theorem 5. Let $X = \text{spec}(A)$ be an affine scheme. We have a bijection

$$\begin{aligned} \{\text{ideals of } A\} &\xrightarrow{\sim} \{\text{closed subschemes of } X\} \\ I \subseteq A &\longmapsto (\text{spec}(A/I), \mathcal{O}_{\text{spec}(A/I)}) \end{aligned}$$

In particular, a closed subscheme of an affine scheme is always affine.

Proof. Let $i : Z \hookrightarrow X$ be a closed subscheme. The forward map is already given as $\alpha : I \mapsto (\text{spec}(A/I), \mathcal{O}_{\text{spec}(A/I)})$. The converse map is proposed to be $\beta : (Z, \mathcal{O}_Z) \mapsto \ker(A \rightarrow \Gamma(Z, \mathcal{O}_Z))$. Let $I_Z = \ker(\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(Z, \mathcal{O}_Z))$. Note that $\Gamma(X, \mathcal{O}_X) = A$ and $i_*\mathcal{O}_Z(X) = \mathcal{O}_Z(i^{-1}(X)) = \mathcal{O}_Z(Z)$. So $I_Z = \ker(f : A \rightarrow \Gamma(Z, \mathcal{O}_Z))$. Our goal is to show $(Z, \mathcal{O}_Z) \simeq (\text{spec}(A/I_Z), \mathcal{O}_{\text{spec}(A/I_Z)})$ because this allows us to see

$$\begin{aligned} \beta \circ \alpha : I_Z &\longmapsto (\text{spec}(A/I_Z), \mathcal{O}_{\text{spec}(A/I_Z)}) \longmapsto \ker(A \rightarrow \Gamma(\text{spec}(A/I_Z), \mathcal{O}_{\text{spec}(A/I_Z)})) \\ \alpha \circ \beta : (Z, \mathcal{O}_Z) &\longmapsto \ker(A \rightarrow \Gamma(Z, \mathcal{O}_Z)) \longmapsto (\text{spec}(A/\ker), \mathcal{O}_{\text{spec}(A/\ker)}) \end{aligned}$$

are identity maps. We first prove $Z \simeq \text{spec}(A/I_Z)$ and then we prove $\mathcal{O}_Z \simeq \mathcal{O}_{\text{spec}(A/I_Z)}$. Consider diagrams

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & \Gamma(Z, \mathcal{O}_Z) & \xrightarrow{\quad} & X \\
 \downarrow & \searrow \chi & & \nearrow & \downarrow \\
 A/I_Z & & & & \text{spec}(A/I_Z)
 \end{array}
 \quad
 \begin{array}{ccc}
 X & \xleftarrow{\text{spec}(f)} & Z \\
 \uparrow & \nwarrow \text{spec}(\chi) & \\
 \text{spec}(A/I_Z) & &
 \end{array}$$

From Theorem 4, we get a Γ -spec adjunction, which provides the existence of the map $Z \rightarrow \text{spec}(A/I_Z)$. Note that $\text{spec}(\Gamma(Z, \mathcal{O}_Z))$ need not to be Z because we cannot invoke Proposition 9 if Z is not affine. Instead, the second diagram is induced by taking spec on all maps. It follows that the second map factors through $\text{spec}(A/I_Z)$.

Without loss of generality, we can assume f is injective. If not, we can mod it out by its kernel. So under this assumption, A/I_Z is A and $\text{spec}(\chi)$ is reduced to the inclusion $Z \hookrightarrow \text{spec}(A) = X$. By definition, the sheaf homomorphism $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z$ is surjective. It remains to show $Z \hookrightarrow X$ is surjective before claiming it to be a homeomorphism. Since $Z \hookrightarrow X$ is already an inclusion, we need to prove there is *not* a proper closed subset of X which contains Z . Let $s \in A$ be such that $Z \subseteq V(s)$.⁴ Let $U \subseteq Z$ be an affine open subset. Consider $f(s)|_U \in \Gamma(U, \mathcal{O}_Z|_U)$ as the image of $f(s)$ under the restriction $\rho : \Gamma(Z, \mathcal{O}_Z) \rightarrow \Gamma(U, \mathcal{O}_Z|_U)$. With enough set-theoretic gymnastics, we can conclude that the vanishing locus of $f(s)|_U$, denoted as $V_U(f(s)|_U)$, is the same as $V(s) \cap U$. Since $U \subseteq Z \subseteq V(s)$, $V_U(f(s)|_U) = U$. It follows that $f(s)|_U \in \Gamma(U, \mathcal{O}_Z|_U)$ is nilpotent. Since $\text{spec}(A)$ is quasi-compact, a closed subset $Z \subseteq \text{spec}(A)$ is quasi-compact as well. This implies that we can cover Z by finitely many such opens. So $f(s)$ is nilpotent. Since f is injective, s is nilpotent. Conclude that $V(s) = \text{spec}(A)$. Indeed, there is not a proper subset of $\text{spec}(A)$ containing Z . As a result, $Z = \text{spec}(A)$.

Next, we check the sheaf isomorphism $\mathcal{O}_X \simeq \mathcal{O}_Z$. Recall that $f : A \rightarrow \Gamma(Z, \mathcal{O}_Z)$ and part of the reduction was f being injective... $\square$

Example (silly). Let $X = \text{spec}(A)$ and $I \subseteq A$ an ideal. $(\text{spec}(A/I), \mathcal{O}_{\text{spec}(A/I)})$ is a closed subscheme of $(X, \mathcal{O}_X)$.

⁴We can at least take $s = 0$.

9/26

Proposition 14. A scheme X is integral if and only if it is irreducible and reduced. Recall that X is integral if $\mathcal{O}_X(U)$ is an integral domain for any open $U \subseteq X$; X is reduced if $\mathcal{O}_X(U)$ has no nilpotent elements.

Proof. ($\Rightarrow$) If X is not irreducible, then we can write $X = X_1 \cup X_2$, where $X_i \subsetneq X$ is a proper closed subset. Then $U_i := X_i^c$ are open and disjoint. We now have an open set $U = U_1 \cup U_2 \subseteq X$ with

$$\mathcal{O}_X(U) \simeq \mathcal{O}_X(U_1 \cup U_2) \simeq \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$$

induced by restriction maps. Note that this is the product of two nonzero rings. So it has zero divisors. It follows that X is not integral. If X is not reduced, it cannot be integral since every integral domain has no nonzero nilpotents. ($\Leftarrow$) Suppose X is both irreducible and reduced. Let $U \subseteq X$ be an open subset. Let $f, g \in \mathcal{O}_X(U)$ such that $fg = 0$. Define

$$U_1 = \{x \in U \mid f_x \in \mathfrak{m}_x\} \quad U_2 = \{x \in U \mid g_x \in \mathfrak{m}_x\}$$

Note that $\mathfrak{m}_x$ is the maximal ideal in $\mathcal{O}_{X,x}$ and $f_x g_x \in \mathfrak{m}_x$ for all $x \dots$ [incomplete] $\square$

Q. If $U, V \subseteq X$ are subschemes, then how to give $U \cap V$ a scheme structure?

Q. Given a map $f : Y \rightarrow X$ and $W \subseteq X$ (or $x \in X$). How to give $f^{-1}(W)$ (or $f^{-1}(x)$) a scheme structure?

A. Fiber product.

Vocabulary 5.3 (fiber product). Given morphisms of schemes $f : X \rightarrow S$ and $g : Y \rightarrow S$. The fiber product of X and Y is $X \times_S Y$ with two projection maps $\text{pr}_X : X \times_S Y \rightarrow X$ and $\text{pr}_Y : X \times_S Y \rightarrow Y$ such that this commutes:

In other words, if $(x, y) \in X \times_S Y$, then $f(x) = g(y) \in S$. It is also universal⁵. In this context, universal means

$$\{T \rightarrow X \times_S Y\} \simeq \left\{ \begin{array}{l} \text{pair of maps } T \rightarrow X, T \rightarrow Y \text{ s.t.} \\ T \rightarrow X \rightarrow S \text{ equals } T \rightarrow Y \rightarrow S \end{array} \right\}$$

which represents a functor $T \longmapsto \text{hom}_{\text{Sch}}(T, X) \times_{\text{hom}_{\text{Sch}}(T, S)} \text{hom}_{\text{Sch}}(T, Y)$.

Proposition 15. If $\text{spec}(A) \rightarrow \text{spec}(R) \leftarrow \text{spec}(B)$ are maps of affine schemes, then $\text{spec}(A \otimes_R B)$ with maps $p : \text{spec}(A \otimes_R B) \rightarrow \text{spec}(A)$ and $q : \text{spec}(A \otimes_R B) \rightarrow \text{spec}(B)$ induced by $A \rightarrow A \otimes_R B \leftarrow B$ is the fiber product.

Proof. For any scheme T , we have

$$\begin{aligned} \text{hom}_{\text{Sch}}(T, \text{spec}(A \otimes_R B)) &\stackrel{\text{thm ??}}{\simeq} \text{hom}_{\text{Ring}}(A \otimes_R B, \mathcal{O}_T(T)) \\ &\stackrel{\alpha \otimes \beta}{\simeq} \text{hom}_{\text{Ring}}(A, \mathcal{O}_T(T)) \times_{\text{hom}_{\text{Ring}}(R, \mathcal{O}_T(T))} \text{hom}_{\text{Ring}}(B, \mathcal{O}_T(T)) \\ &\stackrel{\text{thm ??}}{\simeq} \text{hom}_{\text{Sch}}(T, \text{spec}(A)) \times_{\text{hom}_{\text{Sch}}(T, \text{spec}(R))} \text{hom}_{\text{Sch}}(T, \text{spec}(B)) \end{aligned}$$

where $\alpha : A \rightarrow A \otimes_R B$ is given by $\alpha(a) = a \otimes 1$ and $\beta : B \rightarrow A \otimes_R B$ is given by $\beta(b) = 1 \otimes b$. These two maps induced the second bijection above via composition. $\square$

Remark. The "tensor product of rings" here is in fact a tensor product of R -algebras. And of course $A \otimes_R B$ is an R -algebra by $r(a \otimes b) = (ra \otimes b) = (a \otimes rb)$. Recall that $A \otimes_R B$ being an R -algebra means it is not only a ring, but also a module over R , where the ring addition and the module addition coincide. Multiplication in the ring is given by $(a \otimes b) \cdot (a' \otimes b') = (aa' \otimes bb')$.

Theorem 6. For any pair of maps $X \rightarrow S \leftarrow Y$ of schemes, the fiber product $X \times_S Y$ exists in Sch.

⁵<https://stacks.math.columbia.edu/tag/001V>

6 Week 6 (empty)

7 Week 7 (empty)

8 Week 8 (empty)

9 Week 9 (empty)

10 Week 10

10/27

...

Vocabulary 10.1 ($\mathcal{O}_X$ -module). Let $(X, \mathcal{O}_X)$ be a locally ringed space. An $\mathcal{O}_X$ -module is a sheaf $\mathcal{F}$ on X such that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module for any open $U \subseteq X$. And the restriction maps $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$ are compatible with modules structures via $\mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U)$.

Vocabulary 10.2 ($\mathcal{O}_X$ -module morphism). Given a morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves, Proposition 1 shows that f is injective/bijective if and only if $f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective/bijective for any open U . Now let f be an $\mathcal{O}_X$ -module homomorphism. The kernel is given by the evaluation

$$U \mapsto \ker(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

This is a sheaf denoted by $\ker(f)$. The image is given by

$$a(U \mapsto \ker(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)))$$

Remember that a is sheafification from Proposition 2. It is denoted as $\text{im}(f)$. The cokernel is given by

$$a\left(U \mapsto \mathcal{G}(U) / \text{im}(f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)) = \text{coker}(f_U)\right)$$

It is denoted as $\text{coker}(f)$. These are all $\mathcal{O}_X$ -modules. On stalks $f_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$,

$$\ker(f)_x = \ker(f_x) \quad \text{im}(f)_x = \text{im}(f_x) \quad \text{coker}(f)_x = \text{coker}(f_x)$$

10/29

Vocabulary 10.3 (hom of $\mathcal{O}_X$ -module morphisms). Let $\mathcal{F}, \mathcal{F}'$ be $\mathcal{O}_X$ -modules. The set of $\mathcal{O}_X$ -module morphisms $\text{hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F}')$ is, again, an $\mathcal{O}_X$ -module. Given morphisms f, g and $a \in \mathcal{O}_X(X)$,

$$(f + g)_U := f_U + g_U : \mathcal{F}(U) \rightarrow \mathcal{F}'(U) \quad (af)_U := a|_U f_U$$

The assignment $U \mapsto \text{hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{F}'|_U)$ forms a presheaf. It turned out, this is a sheaf and also an $\mathcal{O}_X$ -module. Write $\text{hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F}')$ for this sheaf.

Vocabulary 10.4 ($\mathcal{O}_X$ -module products). For a set I and a family of $\mathcal{O}_X$ -modules $\mathcal{F}_i$, we consider the assignments

$$a\left(U \mapsto \bigoplus_{i \in I} \mathcal{F}_i(U)\right) \quad U \mapsto \prod_{i \in I} \mathcal{F}_i(U) = \left(\prod_{i \in I} \mathcal{F}_i\right)(U)$$

The restriction maps are induced by restriction maps of the $\mathcal{F}_i$. Addition and scalar multiplication are defined component-wise. Denote the direct sum as $\bigoplus_i \mathcal{F}_i$ and denote the product as $\prod_i \mathcal{F}_i$. Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}_X$ -modules. Their tensor product is defined as

$$a(U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U))$$

This is denoted as $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$. By $f \otimes g$, we mean the image of $f \otimes g \in \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$ in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})(U)$.

Remark. Let $x \in X$ be any point. Any open neighborhood U of x has a canonical map $\mathcal{F}_i(U) \rightarrow \mathcal{F}_{i,x}$ that sends a section s_i to the germ $s_{i,x}$. This immediately gives a map $(\bigoplus_i \mathcal{F}_i)(U) \rightarrow \bigoplus_i \mathcal{F}_{i,x}$. Taking the colimit over all U ,

$$\left(\bigoplus_{i \in I} \mathcal{F}_i\right)_x \simeq \bigoplus_{i \in I} \mathcal{F}_{i,x} \quad (\S)$$

because colimits commute with direct sums. They are isomorphic as $\mathcal{O}_{X,x}$ -modules.

Observation. Note that

$$\mathrm{hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) \simeq \Gamma(X, \mathcal{F}) = \mathcal{F}(X)$$

given by $w \mapsto w_X(1)$ is an isomorphism. Each w has $w_U : \mathcal{O}_X(U) \rightarrow \mathcal{F}(U)$ for $U \subseteq X$ open. So for any $r \in \mathcal{O}_X(U)$ in the ring, $w_U(r) = rw_U(1)$. Since w is a sheaf morphism, each w_U must be the restriction from w_X . In other words, $\rho_{XU}(w_X(1)) = w_U(1)$. If $v_X(1) = w_X(1)$, then $v_U(r) = w_U(r)$ for every $r \in \mathcal{O}_X(U)$ and every open $U \subseteq X$. This would imply $w = v$. Conclude that this map is injective. Given a global section $s \in \Gamma(X, \mathcal{F}) = \mathcal{F}(X)$, define a morphism of sheaves $w : \mathcal{O}_X \rightarrow \mathcal{F}$ by

$$w_U : \mathcal{O}_X(U) \rightarrow \mathcal{F}(U) \quad r \mapsto r(s|_U)$$

on each open $U \subseteq X$. So $w_U(1) = s|_U \in \Gamma(X, \mathcal{F})$ and $w_X(1) = s$. This proves surjectivity. Therefore, the global sections of $\mathcal{F}$ correspond to $\mathcal{O}_X$ -module maps $\mathcal{O}_X \rightarrow \mathcal{F}$.

Vocabulary 10.5 (global section generation). $\mathcal{F}$ is generated by global sections $s_i \in \Gamma(X, \mathcal{F})$ if the corresponding map $(s_i) : \bigoplus_I \mathcal{O}_X \rightarrow \mathcal{F}$ is surjective as sheaves.

Vocabulary 10.6 (locally free). An $\mathcal{O}_X$ -module $\mathcal{F}$ is locally free if for each $x \in X$, there is a neighborhood U of x and a set I such that $\mathcal{F}|_U \simeq \bigoplus_I \mathcal{O}_X|_U$ as $\mathcal{O}_U$ -modules.

Vocabulary 10.7 (rank). If $\mathcal{F}|_U \simeq \bigoplus_I \mathcal{O}_X|_U$, then by (8), $\mathcal{F}_x \simeq \bigoplus_I \mathcal{O}_{X,x}$. The rank of $\mathcal{F}$ at x is $|I|$. Denote this quantity by $\mathrm{rk}_x(\mathcal{F})$. The locally constant function $x \mapsto \mathrm{rk}_x(\mathcal{F})$ is denoted as $\mathrm{rk}(\mathcal{F})$.

Vocabulary 10.8 (invertible sheaf). An invertible sheaf is locally free of constant rank 1.

Vocabulary 10.9 (sheaf of ideal). A sheaf of ideal $\mathcal{I}$ is a sub- $\mathcal{O}_X$ -module $\mathcal{I}(U) \subseteq \mathcal{O}_X(U)$ for each open U .

Vocabulary 10.10 (direct/inverse image of $\mathcal{O}_X$ -module). Suppose $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a locally ringed space morphism. If $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then $f_*\mathcal{F}$ is a $f_*\mathcal{O}_X$ -module. This is because for any open $V \subseteq Y$, $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$ is an $\mathcal{O}_X(f^{-1}(V)) = f_*\mathcal{O}_X(V)$ -module. It is an $\mathcal{O}_Y$ -module via $f^\flat : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. Suppose $s \in \mathcal{O}_Y$ and $m \in f_*\mathcal{O}_X$. The operation is given by $sm := f^\flat(s)m$. This is called the direct image of $\mathcal{F}$. On the other hand, let $\mathcal{G}$ be an $\mathcal{O}_Y$ -module. Also let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on Y . We had

$$f_{\mathrm{pre}}^{-1}(\mathcal{F} \times \mathcal{G})(V) = \mathrm{colim}_{f(V) \subseteq U} \mathcal{F}(U) \times \mathcal{G}(U) \stackrel{(\otimes)}{=} \mathrm{colim}_{f(V) \subseteq U} \mathcal{F}(U) \times \mathrm{colim}_{f(V) \subseteq U} \mathcal{G}(U) = f_{\mathrm{pre}}^{-1}\mathcal{F}(V) \times f_{\mathrm{pre}}^{-1}\mathcal{G}(V)$$

The indexing set $\{U | f(V) \subseteq U\}$ is a set of opens in Y . If $U_1, U_2 \in \{U\}$, then $U_1 \cap U_2 \in \{U\}$. It is also a poset with set inclusion. So $\{U\}$ is filtered and commutes with finite products⁶. This justifies (⊗). It follows that $f^{-1}(\mathcal{F} \times \mathcal{G}) = f^{-1}(\mathcal{F}) \times f^{-1}(\mathcal{G})$. Now let $\mathcal{F}$ be an $\mathcal{O}_Y$ -module. Addition and multiplication are defined by

$$\begin{aligned} f^{-1}(\mathcal{F}) \times f^{-1}(\mathcal{F}) &= f^{-1}(\mathcal{F} \times \mathcal{F}) \longrightarrow f^{-1}(\mathcal{F}) & (a, b) &\longmapsto a + b \\ f^{-1}(\mathcal{O}_Y) \times f^{-1}(\mathcal{F}) &= f^{-1}(\mathcal{O}_Y \times \mathcal{F}) \longrightarrow f^{-1}(\mathcal{F}) & (y, a) &\longmapsto ya \end{aligned}$$

Scalar multiplication makes sense because $\mathcal{F}$ is an $\mathcal{O}_Y$ -module with $\mathcal{O}_Y \times \mathcal{F} \rightarrow \mathcal{F}$. This makes $f^{-1}(\mathcal{F})$ an $f^{-1}(\mathcal{O}_Y)$ -module. By $f^\sharp : f^{-1}(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$, we can make $\mathcal{O}_X$ a $f^{-1}(\mathcal{O}_Y)$ -module: if $m \in \mathcal{O}_X$ and $n \in f^{-1}(\mathcal{O}_Y)$, then $nm := f^\sharp(n)m$. Therefore, it makes sense to define

$$f^*\mathcal{G} := \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{G}$$

Proposition 16. For every $\mathcal{O}_X$ -module $\mathcal{F}$ and every $\mathcal{O}_Y$ -module $\mathcal{G}$ there is an isomorphism of $\Gamma(Y, \mathcal{O}_Y)$ -modules

$$\mathrm{hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \simeq \mathrm{hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

Proof. We check this in steps. Suppose we have a map $\varphi : \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{G} \rightarrow \mathcal{F}$ and $\psi : f^{-1}\mathcal{G} \rightarrow \mathcal{F}$. Define

$$\begin{aligned} \alpha(\varphi) : f^{-1}\mathcal{G} &\rightarrow \mathcal{F} & t &\mapsto \varphi(1 \otimes t) \\ \beta(\varphi) : f^*\mathcal{G} &\rightarrow \mathcal{F} & a \otimes b &\mapsto a \cdot \psi(b) \end{aligned}$$

They are obviously inverses of each other. With these maps and Proposition 3, we have a series of isomorphisms

$$\mathrm{hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \simeq \mathrm{hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F}) \simeq \mathrm{hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

□

⁶<https://stacks.math.columbia.edu/tag/04AX>

Vocabulary 10.11 (attached module on $\text{spec}(A)$). Let A be a ring. Let $X = \text{spec}(A)$ and M be an A -module. Define an $\mathcal{O}_X$ -module $\widetilde{M}$ associated to M as

$$\widetilde{M}(D(f)) = M_f = M \otimes_A A_f$$

for $f \in A$. Following the similar construction in 3.7, we conclude that this is well-defined and $\widetilde{M}$ is a presheaf over the basis $\{D(f) | f \in A\}$. By a similar justification as Theorem 2, $\widetilde{M}$ defined a sheaf on $\{D(f) | f \in A\}$. We say $\widetilde{M}$ is the attached sheaf on X . It turned out $\widetilde{M}$ has the structure of an $\mathcal{O}_X$ -module on $D(f)$ via

$$\begin{aligned} \mathcal{O}_X(D(f)) \times \widetilde{M}(D(f)) &\longrightarrow \widetilde{M}(D(f)) \\ A_f \times M_f &\longrightarrow M_f \end{aligned}$$

This is a module structure map. If $\psi : M \rightarrow N$ is an A -module map, then it induces an A_f -module map $\psi_f : M_f \rightarrow N_f$. For any $f, g \in A$ with $D(f) \subseteq D(g)$, the diagram

$$\begin{array}{ccc} M_g & \xrightarrow{\psi_g} & N_g \\ \downarrow & & \downarrow \\ M_f & \xrightarrow{\psi_f} & N_f \end{array}$$

commutes. So we have $\widetilde{\psi} : \widetilde{M} \rightarrow \widetilde{N}$ as a map of $\mathcal{O}_X$ -modules. This gives a functor $\widetilde{} : A\text{-Mod} \rightarrow \mathcal{O}_X\text{-Mod}$. Let $\mathfrak{p}_x \subseteq A$ be the prime ideal corresponding to $x \in X = \text{spec}(A)$. We have

$$\widetilde{M}_x = \text{colim}_{x \in D(f)} \widetilde{M}(D(f)) = \text{colim}_{f \notin \mathfrak{p}_x} M_f = M_{\mathfrak{p}_x} \quad (\P)$$

10/31

Proposition 17. Let A be a ring, $X = \text{spec}(A)$. For all A -modules M and N , we get mutually inverse bijections

$$\text{hom}_A(M, N) \xrightleftharpoons[\Gamma]{\widetilde{}} \text{hom}_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N})$$

Proof. Start with $\varphi : M \rightarrow N$. Under $\widetilde{}$, this is $\widetilde{\varphi} : \widetilde{M} \rightarrow \widetilde{N}$. Applying Γ gives $\Gamma(\widetilde{\varphi}) : \widetilde{M}(D(f)) = M_f \rightarrow \widetilde{N}(D(f)) = N_f$. This is just φ_f , which gives back $\varphi \in \text{hom}_A(M, N)$. So $\Gamma \circ \widetilde{} = \text{id}$. Now we need to show $\widetilde{}$ is surjective. Let $w : \widetilde{M} \rightarrow \widetilde{N}$ be an $\mathcal{O}_X$ -module map. Define $\varphi = \Gamma(w) = w_X : \widetilde{M}(X) = M \rightarrow \widetilde{N}(X) = N$. We want to show $\widetilde{\varphi} = w$. Note that $w_{D(f)} : \widetilde{M}(D(f)) = M_f \rightarrow \widetilde{N}(D(f)) = N_f$ is the map φ_f . It follows that $\widetilde{\varphi}_{D(f)} = w_{D(f)}$ on every open set. Therefore, $\widetilde{\varphi} = \widetilde{w}_X = w$. $\square$

Corollary. The functor $\widetilde{} : A\text{-Mod} \rightarrow \mathcal{O}_X\text{-Mod}$ is fully faithful. $\square$

Proposition 18. Some quick facts about the attached module $\widetilde{M}$:

1. A sequence of A -modules $M \rightarrow N \rightarrow P$ is exact if and only if $\widetilde{M} \rightarrow \widetilde{N} \rightarrow \widetilde{P}$ an exact sequence of $\mathcal{O}_X$ -modules.
2. Let $f : M \rightarrow N$ be a map of A -modules. We have

$$\ker(\widetilde{f}) = \widetilde{\ker(f)} \quad \text{im}(\widetilde{f}) = \widetilde{\text{im}(f)} \quad \text{coker}(\widetilde{f}) = \widetilde{\text{coker}(f)}$$

3. $\bigoplus_{i \in I} \widetilde{M}_i = \widetilde{\bigoplus_{i \in I} M_i}$ and $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} = \widetilde{M \otimes_A N}$.
4. If $M = \text{colim}_{i \in I} M_i$ is a filtered colimit, then $\widetilde{M} = \text{colim}_i \widetilde{M}_i$.
5. Given $\psi : A \rightarrow B$ with $\varphi : X = \text{spec}(B) \rightarrow Y = \text{spec}(A)$, we have $\varphi^* \widetilde{M} = \widetilde{M \otimes_A B}$ and $\varphi_* \widetilde{N} = {}_A \widetilde{N}$. Here N is viewed as an A -module via ψ .

Proof. (1.) The sequence is exact if and only if the induced sequence $M_{\mathfrak{p}_x} \rightarrow N_{\mathfrak{p}_x} \rightarrow P_{\mathfrak{p}_x}$ is exact. By $(\P)$, this is equivalent to the exactness of $\widetilde{M}_x \rightarrow \widetilde{N}_x \rightarrow \widetilde{P}_x$. Since localization preserves exactness, we obtain (1.) and (2.) follows. (3.) Let $M = \bigoplus_i M_i$. Define inclusions $M_i \hookrightarrow M$. These maps give a homomorphisms $\widetilde{M}_i \hookrightarrow \widetilde{M}$. It follows that $\bigoplus_i \widetilde{M}_i \rightarrow \widetilde{M}$ is a homomorphism. Since localization commutes with direct sums, this map is bijective on all stalks. Therefore, this is an isomorphism. For the second assertion, note that $\widetilde{A} = \mathcal{O}_X$. It is attached to

$$D(f) \mapsto \widetilde{M}(D(f)) \otimes_{\widetilde{A}(D(f))} \widetilde{N}(D(f)) = M_f \otimes_{A_f} N_f \simeq (M \otimes_A N)_f = \widetilde{M \otimes_A N}(D(f))$$

These isomorphisms are compatible with restrictions. (4.) This is a more general version of the direct sum. So same argument works. (5.) Let $f \in A$. Unwind the definition of $\varphi_* \tilde{N}(D(f)) = \tilde{N}(\varphi^{-1}(D(f)))$. For any $x \in X$, $\varphi^{-1}(D(f))$ is the set where $f(\varphi(x)) \neq 0$. This is just $f \circ \varphi = \psi(f) \neq 0$. So $\varphi^{-1}(D(f)) = D(\psi(f))$ and

$$\varphi_* \tilde{N}(D(f)) = \tilde{N}(D(\psi(f))) = N_{\psi(f)}$$

as an A_f -module. This shows $\varphi_* \tilde{N} = {}_A \tilde{N}$ as an A -module. The first part is given by

$$\begin{aligned} \text{hom}_{\mathcal{O}_X}(\varphi^* \tilde{M}, \tilde{W}) &\stackrel{\text{prop ??}}{\simeq} \text{hom}_{\mathcal{O}_Y}(\tilde{M}, \varphi_* \tilde{W}) \simeq \text{hom}_{\mathcal{O}_Y}(\tilde{M}, {}_A \tilde{W}) \\ &\stackrel{\text{prop ??}}{\simeq} \text{hom}_A(M, {}_A W) \stackrel{(!)}{\simeq} \text{hom}_B(M \otimes_A B, W) \stackrel{\text{prop ??}}{\simeq} \text{hom}_{\mathcal{O}_X}(\widetilde{M \otimes_A B}, \tilde{W}) \end{aligned}$$

The (!) isomorphism is due to the change of rings formula. Finally, by Yoneda lemma, $\varphi^* \tilde{M} = \widetilde{M \otimes_A B}$. $\square$

Vocabulary 10.12 (quasi-coherent). Let $(X, \mathcal{O}_X)$ be a locally ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is quasi-coherent if for each $x \in X$, there exists a neighborhood U of x such that there is an exact sequence of $\mathcal{O}_X$ -modules

$$\mathcal{O}_X^{\oplus J}|_U \rightarrow \mathcal{O}_X^{\oplus I}|_U \rightarrow \mathcal{F}|_U \rightarrow 0$$

where I, J are indexing sets. If X is a noetherian scheme, then an $\mathcal{O}_X$ -module $\mathcal{F}$ is coherent if it is quasi-coherent and I, J can be chosen to be finite.

Remark. Let A be a ring and $X = \text{spec}(A)$. Let $M \in A\text{-Mod}$ and $\tilde{M} \in \mathcal{O}_X\text{-Mod}$. And $\tilde{M}|_U$ is an $\mathcal{O}_X|_U$ -module. But $\tilde{M}|_U$ is always free-presentable. In other words, it is the cokernel of a free module homomorphism. So

$$\mathcal{O}_X^{\oplus J}|_U \rightarrow \mathcal{O}_X^{\oplus I}|_U \rightarrow \tilde{M}|_U \rightarrow 0$$

always exists (and exact) in this case.

Theorem 7. Let X be a scheme and $\mathcal{F}$ an $\mathcal{O}_X$ -module. The following statements are equivalent.

1. For every affine open $U = \text{spec}(A) \subseteq X$, there is an A -module M such that $\mathcal{F}|_U \simeq M$ as $\mathcal{O}_U$ -modules.
2. There is an affine open cover $X = \bigcup_i U_i$, where $U_i \simeq \text{spec}(A_i)$, such that for each i , there exists an A_i -module M_i such that $\mathcal{F}|_{U_i} \simeq M_i$ as an $\mathcal{O}_{U_i}$ -module.
3. $\mathcal{F}$ is quasi-coherent.
4. For every open affine subset $U = \text{spec}(A)$ of X and every $f \in A$, there is an isomorphism

$$\Gamma(U, \mathcal{F})_f \simeq \Gamma(D(f), \mathcal{F}) \tag{D}$$

If X is noetherian, the following statements are equivalent.

- 1.' Same as (1.) but with M finitely generated.
- 2.' Same as (2.) but with M finitely generated.
- 3.' $\mathcal{F}$ is coherent.

Proof. (1.) $\Rightarrow$ (2.) is immediate. (2.) $\Rightarrow$ (3.) is immediate. For (3.) $\Rightarrow$ (4.), we may assume $X = \text{spec}(A)$. Now cover $X = \bigcup_i D(g_i)$ with finitely many principal opens such that

$$\tilde{A}_{g_i}^{\oplus J} \xrightarrow{\varphi} \tilde{A}_{g_i}^{\oplus I} \longrightarrow \mathcal{F}|_{D(g_i)} \longrightarrow 0$$

By Proposition ??, we may remove the $\sim$ and obtain $\tilde{\chi} = \varphi$ for some $\chi : A_{g_i}^{\oplus J} \rightarrow A_{g_i}^{\oplus I}$. So $\mathcal{F}|_{D(g_i)} \simeq \text{coker}(\varphi) = \text{coker}(\chi)$. In particular, $\mathcal{F}|_{D(g_i)} \simeq \text{coker}(\chi)$ for an A_{g_i} -module $\text{coker}(\chi)$. But by 10.11, the right hand side is just $\Gamma(D(f), \mathcal{F}|_{D(g_i)}) = \mathcal{F}|_{D(g_i)}(D(f)) = \text{coker}(\chi)_f$, where $f \in A_{g_i}$. The left hand side is $\Gamma(\text{spec}(A_{g_i}), \mathcal{F}|_{D(g_i)}) = \mathcal{F}|_{D(g_i)}(D(1))_f = \text{coker}(\chi)_f$. So $\mathcal{F}|_{D(g_i)}$ satisfies (D). Same argument can show $\mathcal{F}|_{D(g_i g_j)} = \mathcal{F}|_{D(g_i) \cap D(g_j)}$ satisfies (D) as well. Now we go back to the global case. Since $\mathcal{F}$ is a sheaf and localizations are exact, we have

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F})_f & \longrightarrow & \prod_i \Gamma(D(g_i), \mathcal{F})_f & \longrightarrow & \prod_{i,j} \Gamma(D(g_i g_j), \mathcal{F})_f \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \Gamma(D(f), \mathcal{F}) & \longrightarrow & \prod_i \Gamma(D(f g_i), \mathcal{F}) & \longrightarrow & \prod_{i,j} \Gamma(D(f g_i g_j), \mathcal{F}) \end{array}$$

The first row is obtained by localizing the exact sequence from the sheaf condition. It remains exact because localization is exact and commutes with finite products. The local version that we just showed tells us $\mathcal{F}|_{D(g_i)}(D(g_i))_f = \mathcal{F}|_{D(g_i)}(D(f)) = \mathcal{F}(D(g_i) \cap D(f)) = \mathcal{F}(D(fg_i))$. But $\mathcal{F}|_{D(g_i)}(D(g_i))_f = \mathcal{F}(D(g_i))_f$. This shows β and γ are isomorphisms. By the Five Lemma, α is an isomorphism. So $\mathcal{F}$ satisfies (3). To show (4.) $\Rightarrow$ (1.), assume $X = \text{spec}(A) = U$. Set $M = \Gamma(X, \mathcal{F})$. (4.) gives us an isomorphism $\Gamma(D(f), \widetilde{M}) = M_f \simeq \Gamma(D(f), \mathcal{F})$ for all $f \in A$ that are compatible with restriction to $D(g) \subseteq D(f)$. This shows $\mathcal{F} \simeq \widetilde{M}$. $\square$

11 Week 11

11/3

Proposition 19. Let A be a ring and $X = \text{spec}(A)$.

$$\widetilde{(\)} : A\text{-Mod} \xrightleftharpoons{\quad} \text{Qcoh}(\mathcal{O}_X\text{-Mod}) : \Gamma$$

are inverse equivalences. If A is noetherian, then

$$\widetilde{(\)} : A\text{-Mod}^{\text{f.g.}} \xrightleftharpoons{\quad} \text{Coh}(\mathcal{O}_X\text{-Mod}) : \Gamma$$

are inverse equivalences.

Proof. This follows from Theorem ??.

Vocabulary 11.1 (quasi-separated). A scheme is quasi-separated if for any two affine open sets U, V , the intersection $U \cap V$ is quasi-compact. A morphism $f : X \rightarrow Y$ is quasi-separated if $f^{-1}(V)$ is quasi-separated for any affine open $V \subseteq Y$.

Example. Some examples of quasi-separated scheme.

1. If $f : X \rightarrow S$ is separated, then f is quasi-separated. Let $S = \text{spec}(R)$. We claim that if $U = \text{spec}(A)$ and $V = \text{spec}(B)$, then $U \cap V$ is affine. Recall that being separated means $\Delta(X)$ is closed. And we may identify $U \cap V$ as $\{(u, v) | u \in U, v \in V\} \cap \{(x, x)\}$. So we have a map $U \cap V \rightarrow U \times_S V$ and these diagrams

$$\begin{array}{ccccc} U \cap V & \xrightarrow{\quad \varphi \quad} & \Delta^{-1}(U \times_S V) & \xrightarrow{\quad \text{Pr}_{U,V} \quad} & U \times_S V \\ & \searrow & \downarrow & \downarrow & \downarrow \\ & & X & \xrightarrow{\quad \text{Pr}_{1,2} \quad} & X \end{array} \quad \begin{array}{ccccc} \Delta^{-1}(U \times_S V) & \xrightarrow{\quad \psi \quad} & U \cap V & \xrightarrow{\quad} & U \\ & \searrow & \downarrow & \downarrow & \downarrow \\ & & V & \xrightarrow{\quad} & X \end{array}$$

commute. Together, they showed the existence of the dashed morphisms. It turned out φ and ψ are inverses of each other. So $U \cap V = \Delta^{-1}(U \times_S V) = \Delta^{-1}(\text{spec}(A \otimes_R B))$. Since f separated, $\Delta : X \hookrightarrow X \times_S X$ is closed. This implies $U \cap V$ is closed in an affine open. $U \cap V$ is affine as well. Therefore, it is quasi-compact.

2. If X is affine, it is quasi-separated: Let $X = \text{spec}(R)$. An open $U \subseteq X$ is of the form $U = \bigcup_i D(f_i)$ for finitely many i . Similarly, $V = \bigcup_j D(g_j)$. Their intersection is $\bigcup_{i,j} D(f_i g_j)$. It has finite indices as well.
3. If X is locally noetherian, it is quasi-separated: Let A, B be noetherian. $U = \text{spec}(A) \subseteq X$ and $V = \text{spec}(B)$. The intersection $U \cap V \subseteq U = \text{spec}(A)$ is open. Since A is noetherian, U is a noetherian topological space. It follows that every open in U is quasi-compact. So $U \cap V$ is as well.

Vocabulary 11.2 (invertible section). Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. Let $s \in \Gamma(X, \mathcal{L})$. s is invertible at $x \in X$ if there is an open neighborhood U of x such that there is an isomorphism

$$w : \mathcal{L}|_U \simeq \mathcal{O}_U$$

and $w(s_x) \in \mathcal{O}_{X,x}^\times$. Note that since $\mathcal{O}_{X,x}$ is a local ring, $w(s_x) \notin \mathfrak{m}_x$ if and only if s is invertible. There is an induced isomorphism $w_x : \mathcal{L}_x \rightarrow \mathcal{O}_{X,x}$. Now apply this map to the left factor and get

$$\mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} k(x) \simeq \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{X,x}} k(x) \simeq k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$$

So the image of s_x under w is nonzero in $k(x)$ if and only if s is invertible at x . Recall⁷ that if R is a ring, M is an R -module, and $I \subset R$ is an ideal, then $M \otimes_R R/I \simeq M/IM$. Now apply this with $R = \mathcal{O}_{X,x}$, $I = \mathfrak{m}_x$, and $M = \mathcal{L}_x$, we get $\mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} k(x) \simeq \mathcal{L}_x/\mathfrak{m}_x \mathcal{L}_x$. To sum it up,

$$s \text{ is invertible} \iff \bar{s}_x \neq 0 \in \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} k(x) \iff s_x \notin \mathfrak{m}_x \mathcal{L}_x$$

Remark. Write $X_s = \{x \in X | s \text{ invertible}\} \subseteq X$. Note that invertible at x means precisely that its image in the residue field is nonzero. This coincides with $D(s)$ of 3.5. If $w : \mathcal{L}|_U \simeq \mathcal{O}_U$, then $U \cap X_s = X_{w(s)}$ is open.

Theorem 8 (extending sections of quasi-coherent module). Let X be a quasi-coherent, quasi-separated scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. Let $s \in \Gamma(X, \mathcal{L})$ and $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module:

1. Let $t \in \Gamma(X, \mathcal{F})$ such that $t|_{X_s} = 0$. There exists some $n > 0$ with

$$t \otimes s^{\otimes n} = 0 \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n})$$

2. For any $t' \in \Gamma(X_s, \mathcal{F})$, there exists $m > 0$ and $t \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m})$ such that $t|_{X_s} = t' \otimes s^{\otimes m}$.

Proof. Let $X = \bigcup_i U_i$ be a finite union of affine opens such that there is an isomorphism $w_i : \mathcal{L}|_{U_i} \rightarrow \mathcal{O}_{U_i}$ for each i . We fix these isomorphisms.

1. Without loss of generality, can assume X is affine and $\mathcal{L} = \mathcal{O}_X$. By Theorem ??, $\mathcal{F}$ is quasi-coherent if and only if it satisfies $(\mathfrak{D})$. In particular, $\Gamma(X, \mathcal{F})_s \rightarrow \Gamma(D(s), \mathcal{F})$ given by $t_s \mapsto t|_{X_s=D(s)}$ is injective. So $t|_{X_s} = 0$ implies $t_s = 0 \in \Gamma(X, \mathcal{F})_s$. It follows that there exists some $n > 0$ such that $t \otimes s^{\otimes n} = 0 \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n})$.
2. Let s_i be the image of $s|_{U_i}$ in $\mathcal{O}_X(U_i)$ under w_i . Note that $w_i : \mathcal{L}|_{U_i} \simeq \mathcal{O}_{U_i}$ induces $\mathcal{L}^{\otimes n}|_{U_i} \simeq \mathcal{O}_{U_i}$ because $\mathcal{L}|_{U_i}$ is an $\mathcal{O}_{U_i}$ -module. For the same reason, there is an $\mathcal{O}_{U_i}$ -module isomorphism $\mathcal{F}|_{U_i} \otimes_{\mathcal{O}_{U_i}} \mathcal{L}|_{U_i} \simeq \mathcal{F}|_{U_i}$. So $\Gamma(U_i, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) \simeq \Gamma(U_i, \mathcal{F})$. It follows that these statements are the same:

$$t' \in \Gamma(X_s, \mathcal{F}) \iff t' \otimes s^{\otimes \ell} \in \Gamma(X_s, \mathcal{F} \otimes \mathcal{L}^{\otimes \ell}) \text{ for some } \ell \iff (t' \otimes s^{\otimes \ell})|_{X_s \cap U_i} \in \Gamma(X_s \cap U_i, \mathcal{F} \otimes \mathcal{L}^{\otimes \ell})$$

Since $\mathcal{F}$ is quasi-coherent, by Theorem ??, $\Gamma(U_i, \mathcal{F} \otimes \mathcal{L}^{\otimes \ell})_{s_i} \simeq \Gamma(X_s \cap U_i = D(s_i), \mathcal{F} \otimes \mathcal{L}^{\otimes \ell})$. In particular, this is a surjection. So saying $t' \in \Gamma(X_s, \mathcal{F})$ is the same as saying $t_{s_i} \in \Gamma(U_i, \mathcal{F} \otimes \mathcal{L}^{\otimes \ell})_{s_i}$. Clearing denominators, we get $t_i \in \Gamma(U_i, \mathcal{F} \otimes \mathcal{L}^{\otimes \ell})$ for all i . On the open subset $U_i \cap X_s$, we have $t_i|_{X_s \cap U_i} = (t' \otimes s^{\otimes \ell})|_{X_s \cap U_i}$. Now on the triple intersection $X_s \cap U_i \cap U_j$,

$$t_i|_{X_s \cap U_i \cap U_j} = (t' \otimes s^{\otimes \ell})|_{X_s \cap U_i \cap U_j} = t_j|_{X_s \cap U_i \cap U_j} \Rightarrow (t_i|_{U_i \cap U_j} - t_j|_{U_i \cap U_j})|_{X_s \cap U_i \cap U_j} = 0$$

Since X is quasi-separated, $U_i \cap U_j$ is quasi-compact and quasi-separated. By (1.) of this theorem, we can choose some $n > 0$ such that

$$(t_i|_{U_i \cap U_j} - t_j|_{U_i \cap U_j}) \otimes s^n = 0$$

We can choose n to be independent of i and j because they are finite. We first twist ℓ times to extend from X_s to U_i . After that, we twist it extra n times to make them agree on $U_i \cap U_j$ instead of $X_s \cap U_i \cap U_j$, hence glue to a section $t \in \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes(\ell+n)})$ such that $t|_{U_i} = (t_i \otimes s^{\otimes \ell})|_{U_i}$. So $t|_{X_s} = (t' \otimes s^{\otimes(\ell+n)})|_{X_s}$. $\square$

11/5

If $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a map of (locally) ringed spaces, then $f^* : \mathcal{O}_Y\text{-Mod} \rightarrow \mathcal{O}_X\text{-Mod}$ given by $\mathcal{E} \mapsto f^{-1}(\mathcal{E}) \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ is right exact (in fact, exactness is a local property). By 10.10, $f^*(\mathcal{O}_Y) = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{O}_Y \simeq \mathcal{O}_X$. This implies that f^* preserves quasi-coherent modules as

$$\begin{array}{ccccccc} \mathcal{O}_Y^{\oplus J} & \longrightarrow & \mathcal{O}_Y^{\oplus I} & \longrightarrow & \mathcal{F} & \longrightarrow & 0 \\ \downarrow f^* & & \downarrow f^* & & \downarrow f^* & & \\ \mathcal{O}_X^{\oplus J} & \longrightarrow & \mathcal{O}_X^{\oplus I} & \longrightarrow & f^*\mathcal{F} & \longrightarrow & 0 \end{array}$$

In other words, $f^* : \text{Qcoh}(\mathcal{O}_Y\text{-Mod}) \rightarrow \text{Qcoh}(\mathcal{O}_X\text{-Mod})$. If X, Y are noetherian, then f^* preserves coherent modules.

⁷<https://math.stackexchange.com/questions/150114/showing-that-if-r-is-a-commutative-ring-and-m-an-r-module-then-m-otimes>

Vocabulary 11.3 (quasi-compact morphism). Say $f : X \rightarrow Y$ is quasi-compact if $f^{-1}(V)$ is quasi-compact for any affine open $V \subseteq Y$.

Proposition 20. Let $f : X \rightarrow Y$ be quasi-compact and quasi-separated. If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module, then the direct image $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$ -module.

Proof. Can assume Y is affine. It follows that X is quasi-compact and quasi-separated. So we can cover X by finitely many affine opens $X = \bigcup_i U_i$ with $U_i \cap U_j$ being quasi-compact. Cover $U_i \cap U_j$ by finitely many affine set: $U_i \cap U_j = \bigcup_k U_{ijk}$. Now we have an exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & f_*\mathcal{F} & \longrightarrow & \bigoplus_i f_*\mathcal{F}|_{U_i} & \longrightarrow & \bigoplus_{i,j} f_*\mathcal{F}|_{U_i \cap U_j} \\ & & & & & \searrow & \downarrow \\ & & & & & & \bigoplus_{i,j,k} f_*\mathcal{F}|_{U_{ijk}} \end{array}$$

The top row is just the sheaf condition. So it is exact. Since U_{ijk} cover $U_i \cap U_j$, the sequence with ijk is exact as well. This proves that $f_*\mathcal{F}$ is quasi-coherent. $\square$

Vocabulary 11.4 (attached module on $\text{proj}(A)$). Let $A = \bigoplus_{i \geq 0} A_i$ be a graded ring. Let $A_+ = \bigoplus_{i \geq 1} A_i$. Let M be a graded A -module. Define the associated quasi-coherent module $\widetilde{M}$ on $\text{proj}(A)$ as

$$M_{(f)} =: \Gamma(D_+(f), \widetilde{M}) \quad (2)$$

where $f \in A_+$ is homogeneous. If $D_+(g) \subseteq D_+(f)$, then $V_+(f) \subseteq V_+(g)$. This implies that $g \in \text{rad}(\langle f \rangle)_+$. Conclude that $f \in A_g^\times$. This means there is an induced $A_{(f)}$ -module map $M_f \rightarrow M_g$. Hence an $A_{(f)}$ -module map

$$\Gamma(D_+(f), \widetilde{M}) = M_{(f)} \longrightarrow M_{(g)} = \Gamma(D_+(g), \widetilde{M}) \quad (\sigma')$$

There is a unique sheaf satisfying both (2) and (σ') . Let $X = \text{proj}(A)$. We obtain a functor

$$\widetilde{(\)} : \text{Gr}A\text{-Mod} \longrightarrow \text{Qcoh}(\mathcal{O}_X\text{-Mod})$$

If A is noetherian, then

$$\widetilde{(\)} : \text{Gr}A\text{-Mod}^{\text{f.g.}} \longrightarrow \text{Coh}(\mathcal{O}_X\text{-Mod})$$

Assume A is generated by A_1 . This $\widetilde{(\)}$ is exact and commutes with sums, filtered colimits, and tensor products. But it is *not* faithful. Indeed, if M is a graded A -module and $M_i = 0$ for all $i \gg 0$, then $M_{(f)} = 0$ for all homogeneous $f \in A_{d \geq 0}$ because

$$M_{(f)} = \left\{ \frac{m}{f^n} \mid m \text{ homog, } \deg(m) = nd \right\} = \left\{ \frac{mf^k}{f^{n+k}} \mid \deg(mf^k) = \deg(m) + k \deg(f) = nd + kd \right\}$$

If k is large enough, then $M_{(f)} = 0$. So $\widetilde{M} = 0$. The above equation also says $M_{(f)} = (M_f)_0$.

Vocabulary 11.5 (serre's twisting sheaf). Given graded A -module M . Write $M(n)$ where $n \in \mathbb{Z}$ for the shifted module $M(n)_i = M_{n+i}$. Define $\mathcal{O}_X(n) := \widetilde{A(n)}$. For any sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$ we set $\mathcal{F}(n) := \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.

Proposition 21. Let A be a graded ring. Let M be a graded A -module. Assume A_+ is generated by A_1 . Let $X = \text{proj}(A)$. $\mathcal{O}_X(n)$ is an invertible $\mathcal{O}_X$ -module. Moreover,

$$\mathcal{O}_X(1)^{\otimes n} \simeq \mathcal{O}_X(n) \quad \text{and} \quad \widetilde{M}(n) = \widetilde{M(n)}$$

Proof. We need to show $\mathcal{O}_X(n)$ is locally free of rank 1. Since $X = \bigcup_{f \in A_+} D_+(f)$ and A_1 generates A_+ , $X = \bigcup_{f \in A_1} D_+(f)$. So for any $f \in A_1$, $A_{(f)} = (A_f)_0$. On the other hand, $(A_f)_n = (A_f(n))_0 = (A(n)_f)_0 = A(n)_{(f)}$. So

$$A_{(f)} = (A_f)_0 \xrightarrow[(-) \cdot f^k]{\sim} (A_f)_{dk} (A_f)_n = A(n)_{(f)}$$

where d is the grading of f itself and $n = dk$ is a multiple of d . But $\mathcal{O}_X(D_+(f)) = \widetilde{A}(D_+(f)) = A_{(f)}$ and $\mathcal{O}_X(n)(D_+(f)) = \widetilde{A(n)}(D_+(f)) = A(n)_{(f)}$. This gives us a section isomorphism $\mathcal{O}_X(n)(D_+(f)) \simeq \mathcal{O}_X(D_+(f))$ by

$(-) \cdot f^{-k}$. This is the same as saying $\mathcal{O}_X(n)|_{D_+(f)} \simeq \mathcal{O}_X|_{D_+(f)}$, which is the local trivialization of $\mathcal{O}_X(n)$. We have an isomorphism $A(n) \otimes_A A(m) \simeq A(n+m)$ as A -modules given by $a \otimes b \mapsto ab$ and $x \mapsto x \otimes 1$. We conclude

$$\begin{array}{ccccc} & A(n) \otimes_A A(m) & \xrightarrow{\sim} & A(n+m) & \\ \swarrow \widetilde{(-)} & & & & \searrow \widetilde{(-)} \\ A(n) \widetilde{\otimes}_A A(m) & \xrightarrow[\text{prop } ??]{\sim} \widetilde{A(n)} \otimes_{\widetilde{A}} \widetilde{A(m)} & \xrightarrow[\text{11.5}]{\sim} \mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) & \xrightarrow{\varphi} & \mathcal{O}_X(n+m) \end{array}$$

by applying the $\widetilde{(-)}$ functor and previous results. Note that φ is usually not an isomorphism. But in this case, we can find some power i such that $(-) \cdot f^{-i}$ gives a local trivialization $\mathcal{O}_X(m)|_{D_+(f)} \simeq \mathcal{O}_X|_{D_+(f)}$. Similarly, we may find a trivialization for $\mathcal{O}_X(n+m)|_{D_+(f)} \simeq \mathcal{O}_X|_{D_+(f)}$. This makes $\varphi|_{D_+(f)}$ an isomorphism. But we assume that A_+ is generated by A_1 . This implies we may cover X by $D_+(f)$. Since these are sheaves, an isomorphism on each $D_+(f)$ is an isomorphism on all of X . Therefore, φ is an isomorphism. We have previously shown $A(f) \simeq A(n)_{(f)}$. It follows that $M_{(f)} \simeq M(n)_{(f)}$ and thus $A(n)_{(f)} \otimes_{A(f)} M_{(f)} \simeq M(n)_{(f)}$. So

$$\begin{aligned} \mathcal{O}_X(n)(D_+(f)) \otimes_{\mathcal{O}_X(D_+(f))} \widetilde{M}(D_+(f)) &\simeq \widetilde{M(n)}(D_+(f)) \iff \mathcal{O}_X(n)|_{D_+(f)} \otimes_{\mathcal{O}_{D_+(f)}} \widetilde{M}|_{D_+(f)} \simeq \widetilde{M(n)}|_{D_+(f)} \\ &\iff \mathcal{O}_{D_+(f)} \otimes_{\mathcal{O}_{D_+(f)}} \widetilde{M(n)}|_{D_+(f)} \simeq \widetilde{M(n)}|_{D_+(f)} \end{aligned}$$

It suffices to show it is an isomorphism on each $D_+(f)$. Conclude that $\widetilde{M}(n) = \widetilde{M(n)}$. $\square$

Proposition 22 (projection formula). Let $f : X \rightarrow Y$ be a morphism of ringed spaces, $\mathcal{F}$ an $\mathcal{O}_X$ -module and $\mathcal{E}$ a locally free $\mathcal{O}_Y$ -module of finite rank. Then there is a natural isomorphism

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \simeq f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E}$$

Proof. Applying $f^*\mathcal{E}$ to the counit $f^*f_*\mathcal{F} \rightarrow \mathcal{F}$ of the adjunction from Proposition ?? gives

$$f^*(f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E}) \simeq f^*f_*\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}$$

which corresponds to a morphism $g : f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{E} \rightarrow f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E})$. We want to show g is an isomorphism. Since $\mathcal{E}$ is locally free, on any open neighborhood U of $y \in Y$, $\mathcal{E} \simeq \mathcal{O}_Y^{\oplus n}$. Note that both sides of g are composites of additive functors and they commute with finite direct sums⁸. So without loss of generality, we can assume $\mathcal{E} \simeq \mathcal{O}_Y$:

$$f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y \simeq f_*\mathcal{F} \simeq f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X) \simeq f_*(\mathcal{F} \otimes_{\mathcal{O}_X} (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{O}_Y)) \simeq f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{O}_Y)$$

By the adjunction from Proposition ??, the isomorphism above $f_*\mathcal{F} \simeq f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{O}_Y)$ corresponds to an isomorphism $f^*f_*\mathcal{F} \simeq f^*(f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y) \simeq \mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{O}_Y$. Applying the adjunction one more time, we conclude that $g|_U : f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y \rightarrow f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{O}_Y)$ is an isomorphism. $\square$

Corollary. Suppose A is a graded ring such that A_+ is generated by A_1 . Let $\mathcal{F} \in \text{Qcoh}(X)$ and let $\varphi : A \rightarrow B$ be a graded ring map. Define $G(\varphi) := \{q \in \text{proj}(B) | \varphi^{-1}(\mathfrak{p}) \not\subseteq A_+\} \subseteq \text{proj}(B)$. Denote $U := G(\varphi)$ and $Y := \text{proj}(B)$. If there is a map $f : U \rightarrow X = \text{proj}(A)$, then

$$f^*\mathcal{O}_X(n) \simeq \mathcal{O}_Y(n)|_U \quad \text{and} \quad f_*(\mathcal{O}_Y(n)|_U) \simeq (f_*\mathcal{O}_U)(n)$$

Proof idea. This may contain crazy amount of details. But it mainly follows from Proposition ??. $\square$

11/7

Vocabulary 11.6 (associated graded A -module). Let $A = \bigoplus_{n \geq 0} A_n$ be a graded ring, let $X = \text{Proj } A$, and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. The associated graded A -module of $\mathcal{F}$ is

$$\Gamma_*(\mathcal{F}) := \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n)).$$

The A -module structure is defined using the natural maps $\beta_n : A_n \rightarrow \Gamma(X, \mathcal{O}_X(n))$. If $a \in A_n$ and $x \in \Gamma_m(\mathcal{F}) = \Gamma(X, \mathcal{F}(m))$, then $a \cdot x$ is defined to be the image of $\beta_n(a) \otimes x$ under the natural multiplication map $\Gamma(X, \mathcal{O}_X(n)) \otimes \Gamma(X, \mathcal{F}(m)) \rightarrow \Gamma(X, \mathcal{F}(n+m))$, induced locally by the canonical isomorphism $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{F}(m) \simeq \mathcal{F}(n+m)$.

Proposition 23. If $X = \text{proj}(R[x_0, \dots, x_r]) = \mathbb{P}_R^r$, then $\Gamma_*(\mathcal{O}_X) \simeq R[x_0, \dots, x_r]$.

...

⁸<https://stacks.math.columbia.edu/tag/0DLP>

12 Week 12

Vocabulary 12.1 (very ample). Let $f : X \rightarrow Y$ be a map of schemes and let $\mathcal{L}$ be an invertible sheaf on X . We say $\mathcal{L}$ is very ample relative to Y if there is an immersion (open inside of a closed subscheme) $i : X \hookrightarrow \mathbb{P}_Y^n$ over Y such that $i^* \mathcal{O}_{\mathbb{P}_Y^n}(1) \simeq \mathcal{L}$. Write $\mathcal{O}_X(1)$ for $\mathcal{L}$ in this case.

Proposition 24. If $X = \text{spec}(A)$ is an affine scheme, then there is an 1-1 correspondence between ideals $I \subseteq A$ and closed subschemes $Y \subseteq X$ given by $I \mapsto \text{image of } \text{spec}(A/I) \text{ in } X$. In particular, every closed subscheme of an affine scheme is affine.

Proof. Recall that by Proposition ?? we have an equivalence $\widetilde{} : A\text{-Mod} \longleftrightarrow \text{Qcoh}(\mathcal{O}_X\text{-Mod}) : \Gamma$. It follows that the quasi-coherent sheaves of ideals on X are in 1-1 correspondence with ideals of A . $\square$

Proposition 25. Let $f : X \rightarrow Y$ be a morphism of schemes. If f is a closed immersion, then it is a finite morphism.

Proof. Without loss of generality, assume $Y = \text{spec}(A)$. Since closed subschemes correspond to ideals $I \subseteq A$, the closed immersion is $X = \text{spec}(A/I) \hookrightarrow \text{spec}(A)$ by our previous proposition. This corresponds to the ring map $A \rightarrow A/I$. To show finiteness, we need to show A/I is a finitely generated A -module. Indeed, $\langle 1 + I \rangle = A/I$. $\square$

Proposition 26. Let $f : X \rightarrow Y$ be a morphism of schemes. If f is a finite morphism of noetherian schemes, and if $\mathcal{F}$ is coherent on X , then $f_* \mathcal{F}$ is coherent on Y .

Proof. Since $f : X \rightarrow Y$ is finite, for any affine piece $U = \text{spec}(A)$ in Y , $f^{-1}(U) = \text{spec}(B)$ for some finite A -module B . By Theorem ??(2'), there exists some finite B -module M such that $\mathcal{F}|_{f^{-1}U} = \widetilde{M}$. By Proposition ??(5), $f_* \mathcal{F}|_U = {}_A M$. Note that M is finite over A as well. $f_* \mathcal{F}$ is coherent on Y . $\square$

Theorem 9 (Serre). Let X be a projective scheme over a noetherian ring A . Let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X . Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module. Then there exists $n_0 \in \mathbb{Z}$ such that for all $n \geq n_0$, $\mathcal{F}(n)$ is generated by finitely many global sections.

Proof. There is a closed immersion $i : X \hookrightarrow \mathbb{P}_A^r$ such that $\mathcal{O}_X(1) \simeq i^* \mathcal{O}_{\mathbb{P}^r}(1)$. Since i is a closed immersion, it is finite by Proposition ??. And by Proposition ??, $i_* \mathcal{F}$ is coherent on $\mathbb{P}^r$. By the projection formula

$$i_*(\mathcal{F}(n)) \simeq i_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)) \simeq i_*(\mathcal{F} \otimes_{\mathcal{O}_X} i^* \mathcal{O}_{\mathbb{P}^r}(n)) \simeq i_* \mathcal{F} \otimes_{\mathcal{O}_{\mathbb{P}^r}} \mathcal{O}_{\mathbb{P}^r}(n) = (i_* \mathcal{F})(n)$$

Use the observation after 10.4 and the $i^* i_*$ adjunction:

$$\begin{aligned} \Gamma(\mathbb{P}^r, i_* \mathcal{F}(n)) &\simeq \text{hom}_{\mathcal{O}_{\mathbb{P}^r}}(\mathcal{O}_{\mathbb{P}^r}, i_* \mathcal{F}(n)) \simeq \text{hom}_{\mathcal{O}_X}(i^* \mathcal{O}_{\mathbb{P}^r}, \mathcal{F}(n)) \\ &\simeq \text{hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}(n)) \simeq \Gamma(X, \mathcal{F}(n)) \end{aligned}$$

Therefore, it is safe to assume $X = \mathbb{P}^r$. Now we cover $X = \mathbb{P}^r$ with affine patches $D_+(x_i) = \text{spec}(A[x_0/x_i, \dots, x_r/x_i])$. Since $\mathcal{F}$ is coherent, by Theorem ??(2'), there exists a finite module M_i over $A[x_0/x_i, \dots, x_r/x_i]$ such that $\mathcal{F}|_{D_+(x_i)} = \widetilde{M}_i$. For each i , take a finite number of elements $s_{ij} \in M_i$ which generate M_i . Note that we are using the spec version of the attached module because we are on an affine patch. So by 10.11, $s_{ij} \in \Gamma(\text{spec}(A[x_0/x_i, \dots, x_r/x_i]), \widetilde{M}_i) = (M_i)_1 = M_i$. Since x_i is homogeneous of degree 1, $x_i \in \Gamma(X, \mathcal{O}_X(1))$. The nonvanishing locus of the homogeneous coordinate x_i is the principal open $D_+(x_i)$. By Theorem ??(2), there exists an integer $n > 0$ and a section $t_{ij} \in \Gamma(X, \mathcal{F} \otimes \mathcal{O}_X(1)^{\otimes n})$ such that on $D_+(x_i)$ we have $t_{ij}|_{D_+(x_i)} = s_{ij} \otimes x_i^{\otimes n}$. By Proposition ??, $t_{ij} \in \Gamma(X, \mathcal{F}(n))$ and by the argument outlined in the same proposition, multiplication by x_i^n is invertible on $D_+(x_i)$. Therefore, $\mathcal{O}_X(n)|_{D_+(x_i)} \simeq \mathcal{O}_X|_{D_+(x_i)}$. This induces a map

$$\mathcal{F}(n)|_{D_+(x_i)} = \mathcal{F}|_{D_+(x_i)} \otimes \mathcal{O}_X(n)|_{D_+(x_i)} \simeq \mathcal{F}|_{D_+(x_i)} \otimes \mathcal{O}_X|_{D_+(x_i)} = \mathcal{F}|_{D_+(x_i)}$$

Again, by Theorem ??(2), there exists some $A[x_0/x_i, \dots, x_r/x_i]$ -module M'_i such that $\mathcal{F}(n)|_{D_+(x_i)} = \widetilde{M}'_i$. The above isomorphism gives $M_i \simeq M'_i$ by $s_{ij} \mapsto s_{ij} \otimes x_i^{\otimes n}$. Since $\{s_{ij}\}$ generate M_i , we must have $\{s_{ij} \otimes x_i^{\otimes n}\}$ generate M'_i . Since the elements $s_{ij} \otimes x_i^{\otimes n}$ generate the $A[x_0/x_i, \dots, x_r/x_i]$ -module M'_i , the restrictions $t_{ij}|_{D_+(x_i)}$ generate the sheaf $\mathcal{F}(n)|_{D_+(x_i)}$. As the opens $D_+(x_i)$ cover X , the global sections t_{ij} generate $\mathcal{F}(n)$. $\square$

Remark. In the proof above, the restrictions $t_{ij}|_{D_+(x_i)}$ generate the sheaf $\mathcal{F}(n)|_{D_+(x_i)}$ is not automatic. Let $B = A[x_0/x_i, \dots, x_r/x_i]$ and $U = \text{spec}(B) = D_+(x_i)$. And for simplicity, rename $M = M'_i$. Take any $x \in U$. By (9), the stalk of $\widetilde{M}$ at x is $\widetilde{M}_x = M_{\mathfrak{p}_x}$. We knew $\{t_{ij}|_U\}$ generate M . Moreover, every element in $M_{\mathfrak{p}_x}$ can be written as m/s , where $m \in M$ and $s \notin \mathfrak{p}_x$. So for some $b_j \in B$, $m/s = \sum_j (b_j/s)t_{ij}|_U \in M_{\mathfrak{p}_x}$ and $b_j/s \in B_{\mathfrak{p}_x}$. The images of $t_{ij}|_U$ in the localization generate $M_{\mathfrak{p}_x}$. We obtain a map

$$(B_{\mathfrak{p}_x} t_{ij}|_U)^{\oplus N} \simeq (B_{\mathfrak{p}_x})^{\oplus N} \rightarrow M_{\mathfrak{p}_x}$$

This gives us a surjection of sheaves $\mathcal{O}_U^{\oplus N} \rightarrow \widetilde{M}$.

Proposition 27 (useful). Let X be projective over a noetherian ring A . Then any coherent sheaf $\mathcal{F}$ on X can be written as a quotient of a sheaf $\mathcal{E}$, where $\mathcal{E}$ is a finite direct sum of twisted structure sheaves $\mathcal{O}(n_i)$ for various numbers n_i . In other words, we have a surjection

$$\varphi : \mathcal{E} = \bigoplus_{i=1}^N \mathcal{O}(n_i) \rightarrow \mathcal{F}$$

So $\mathcal{F} = \mathcal{E} / \ker(\varphi)$.

Proof. Let $\mathcal{F}(n)$ be generated by finite global sections. This is possible due to Serre's theorem on sheaves generated by sections. Then we have a surjection $\bigoplus_{i=1}^N \mathcal{O}_X \rightarrow \mathcal{F}(n) \rightarrow 0$. Tensoring with $\mathcal{O}_X(-n)$, we get a surjection $\bigoplus_{i=1}^N \mathcal{O}_X(-n) \rightarrow \mathcal{F} \rightarrow 0$ as required. $\square$

13 Appendix

Theorem A1. Let A be a commutative ring. Every idempotent of $A/\text{nilrad}(A)$ lifts to some idempotent of A .

Proof. Let $\bar{x} \in A/\text{nilrad}(A)$. If $x \in A$ is some lift of $\bar{x}$, then $x^2 - x \in \text{nilrad}(A)$ and $x^2 - x$ is nilpotent. So there exists some n such that $(x^2 - x)^n = x^n(x - 1)^n = 0$. Since x^n and $(x - 1)^n$ coprime, Chinese remainder theorem provides an isomorphism

$$\varphi : A \xrightarrow{\sim} A/\langle x^n \rangle \times A/\langle (x - 1)^n \rangle \quad a \mapsto (a \bmod x^n, a \bmod (x - 1)^n)$$

Define $e = \varphi^{-1}((0, 1))$. In a product ring like the one above, $(0, 1)^2 = (0, 1)$. So we can conclude that $e^2 = e$ through φ . And also $x - e$ is nilpotent because in each factor it lies in a nilpotent ideal. e is the idempotent lift. $\square$

Index

- $\mathcal{O}_X$ -module, 21
- $\mathcal{O}_X$ -module morphism, 21
- $\mathcal{O}_X$ -module products, 21

- affine algebraic sets, 3
- affine coordinate rings, 3
- affine scheme, 13
- associated graded A -module, 28
- attached module on $\operatorname{spec}(A)$, 23
- attached module on $\operatorname{proj}(A)$, 27

- category of $\mathbf{Psh}(X)$, 4
- closed immersion, 17
- closed subscheme, 17
- connected scheme, 16
- constant sheaf, 7

- direct image, 7
- direct/inverse image of $\mathcal{O}_X$ -module, 22

- fiber product, 19

- generic point, 16
- global section generation, 22

- hom of $\mathcal{O}_X$ -module morphisms, 21

- ideal of subset, 10
- in/sur/bijective morphisms, 6
- inverse image, 7
- invertible section, 25
- invertible sheaf, 22
- irreducible scheme, 16
- irreducible space, 15

- locally free, 22
- locally ringed space, 8

- maps between locally ringed spaces, 9
- morphism of schemes, 13
- morphisms of algebraic sets, 3

- open immersion, 17

- poset of opens, 4
- presheaf, 4
- principal open, 10

- quasi-coherent, 24
- quasi-compact morphism, 27
- quasi-compact scheme, 16
- quasi-separated, 25

- rank, 22
- residue field, 9
- ringed space, 8

- scheme, 13
- section, 13
- serre's twisting sheaf, 27
- sheaf, 4
- sheaf of ideal, 22
- sheaf on spec , 12
- sheafification, 7
- spec , 9
- spec map, 11
- specialization, 16
- stalks, 5

- very ample, 29

- Zariski locus of vanishing, 9
- Zariski topology on spec , 9